

GENERAL NOTES:

1. THE PROJECT SPECIFICATIONS (A BOOK OF SPECIFICATIONS WHEN PROVIDED) ARE A PART OF THE PROJECT DOCUMENTS. IF THERE IS A DISCREPANCY FOUND BETWEEN THE SPECIFICATIONS AND THE DRAWINGS, SPECIFICATIONS TAKE PRECEDENCE, HOWEVER THE MATTER MUST BE PROMPTLY SUBMITTED TO THE DESIGNER OF RECORD FOR CLARIFICATION. ANY WORK PERFORMED BY THE CONTRACTOR WITHOUT SUCH A CLARIFICATION WILL BE AT CONTRACTOR'S OWN RISK AND EXPENSE.
2. EXAMINE THE DRAWINGS AND THE SPECIFICATIONS AND NOTIFY THE ENGINEER & CONTRACTING OFFICER OF ANY DISCREPANCIES IN ELEVATIONS, DIMENSIONS, AND SITE CONDITIONS INCLUDING ERRORS BEFORE PROCEEDING WITH ANY WORK. OMISSIONS AND CONFLICTS BETWEEN THE VARIOUS ELEMENTS OF THE DRAWINGS (AND SPECIFICATIONS) WILL BE RESOLVED IN WRITING WITH THE ENGINEER/ARCHITECT & CONTRACTING OFFICER PRIOR TO START OF WORK.
3. THE DRAWINGS (AND SPECIFICATIONS) REPRESENT THE COMPLETED STRUCTURE. THEY DO NOT INDICATE THE METHOD OF CONSTRUCTION. PROVIDE ALL MEASURES AND MEANS NECESSARY TO PROTECT PERSONS AND THE STRUCTURE DURING CONSTRUCTION. SUCH MEASURES WILL INCLUDE, BUT NOT BE LIMITED TO BRACING, SHORING, ETC. OBSERVATION VISITS BY THE ARCHITECT OR ENGINEER DOES NOT INCLUDE REVIEW OF THESE MEASURES.
4. TYPICAL DETAILS WILL BE USED WHENEVER APPLICABLE WHETHER SPECIFICALLY REFERENCED OR NOT. DRAWINGS WILL NOT BE SCALED FOR CONSTRUCTION PURPOSES.
5. CONDITIONS SHOWN ARE BASED ON EXISTING PLANS AND FIELD OBSERVATION WITHOUT DEMOLITION. AFTER DEMOLITION, AN CLARIFICATION REQUIRED TO DETERMINE SCOPE OF WORK MUST BE BROUGHT TO THE ATTENTION OF THE CONSTRUCTION MANAGER.
6. DIMENSIONS SHOWN ARE AN APPROXIMATE. FIELD VERIFY ALL DIMENSIONS AND SITE CONDITIONS PRIOR TO PROCURING MATERIALS OR COMMENCING WORK.
7. EQUIPMENT, PIPING, WALLS, AND SYSTEMS NOTED FOR DEMOLITION SHALL BE REMOVED AND DISPOSED OF BY THE CONTRACTOR, UNLESS NOTED OTHERWISE. DEMOLITION SHALL INCLUDE ASSOCIATED SUPPORTS, ANCHOR BOLTS AND OTHER APPURTENANCES.
8. WHERE DUCTWORK, PIPING, WIRING, BOLTS, FOUNDATIONS, ETC ARE REMOVED AND NOT REINSTALLED, PATCH ALL HOLES IN WALLS, FLOOR, AND CEILING WITH LIKE MATERIAL.
9. LABEL PIPE SYSTEMS PER THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME) A13.1, SCHEME FOR THE IDENTIFICATION OF PIPING SYSTEMS. CONTACT THE COR FOR ANY DEVIATION FROM THE ASME A13.1 PRIOR TO CONSTRUCTION.
10. ROOFING WORK MUST BE PERFORMED BY A ROOFING CONTRACTOR CERTIFIED BY THE EXISTING ROOF SYSTEM MANUFACTURER. WORK PERFORMED ON THE ROOF MUST NOT VOID THE EXISTING ROOF WARRANTY.
11. DUST GENERATED FROM THE DEMOLITION WORK MUST BE CONTAINED.
12. EQUIPMENT LOADS MUST NOT BE ADDED TO A BUILDING ROOF WITHOUT PRIOR APPROVAL FROM THE CONSTRUCTION MANAGER.
13. PRESSURE SYSTEM WELDS (PIPING, PRESSURE VESSELS, TANKS, HEAT EXCHANGERS, ETC.) MUST BE ACCOMPLISHED BY WELDERS CERTIFIED PER THE MOST RECENT EDITION OF ASME BPVC SEC IX, "WELDING, BRAZING, AND FUSING QUALIFICATIONS".
14. BEFORE BURNING / WELDING / SOLDERING, OBTAIN A PERMIT FROM THE MCAS CHERRY POINT FIRE DEPARTMENT (252-466-2241). A FIRE WATCH SHALL BE MAINTAINED BOTH DURING AND AT LEAST 120 MINUTES AFTER BURNING / WELDING / SOLDERING WORK.
15. OCCUPIED SPACE MUST BE CLEANED AT THE COMPLETION OF EACH WORK PERIOD. REMOVE AND DISPOSE OF SCRAP MATERIALS AND THOROUGHLY CLEAN THE CONSTRUCTION AREA DAILY.
16. DISPOSE OF DEBRIS, RUBBISH, SCRAP, AND OTHER UNSALVAGEABLE MATERIALS IN ACCORDANCE WITH CONTRACT DOCUMENTS. STORAGE AND/OR BURNING OF REMOVED ITEMS IS PROHIBITED.
17. THE CONTRACTOR MUST CORRECT AT HIS OWN EXPENSE, DAMAGES TO WORK OR MATERIALS IN PLACE ON THE PREMISES, OR INCLUDED IN THIS CONTRACT, DURING THE EXECUTION OF THIS CONTRACT.
18. COMPLETE FORM 13700/202 (PLANT EQUIPMENT CHECKLIST) FOR EACH NEW PIECE OR EQUIPMENT AND SEND THE COMPLETED FORMS TO THE CONSTRUCTION MANAGER.
19. PROVIDE EQUIPMENT MANUALS (INCLUDING OPERATIONS, MAINTENANCE, SPARE PARTS LIST, AND ILLUSTRATED PARTS BREAKDOWN) IN ACCORDANCE WITH CONTRACT DOCUMENTS.
20. DO NOT SCALE DRAWINGS FOR SHOP DRAWINGS.

RECYCLING NOTES:

1. DISPOSE OF THE FOLLOWING MATERIALS AT THE FRC EAST RECYCLING FACILITY: STEEL, ALUMINUM, BRASS, COPPER (WIRE), MAGNESIUM, HIGH TEMPERATURE STEEL, COMPOSITE/FIBERGLASS, PLASTIC DRINK BOTTLES, PLASTIC TOOL BOX INSERTS, SCRAP ELECTRONICS, TONER CARTRIDGES, UNPAINTED/UNTREATED LUMBER AND CRATES, CARDBOARD ONLY, X-RAY FILM, KIRKSITE, LEAD, STAINLESS STEEL, AND TIRES. FRC EAST RECYCLING CENTER B4785 IS LOCATED BEHIND B4225.
2. THE FOLLOWING MATERIALS ARE NOT ACCEPTED AT THE FRC EAST RECYCLING FACILITY: WOODEN FURNITURE OF ANY KIND, PACKING MATERIAL, PLASTIC OF ANY KIND (OTHER THAN THOSE MENTIONED IN NOTE 2), PAINTED OR TREATED LUMBER, BOXES CONSISTING OF BOTH CARDBOARD AND WOOD, REFRIGERATORS, PLANT ACCOUNTED EQUIPMENT NOT PROCESSED THROUGH CAN 1375, RUBBER FLOOR MATS, GENERAL REFUSE AND CARDBOARD SHIPPING TUBES.

ENVIRONMENTAL NOTES:

1. NOTIFY THE CONTRACTING OFFICER REPRESENTATIVE WHEN THE PROJECT INCLUDES INSTALLATION OF EQUIPMENT THAT MAY REQUIRE AN AIR PERMIT OR REMOVAL OF EQUIPMENT CURRENTLY LISTED ON THE EXISTING FRC EAST AIR PERMIT. NOTIFICATION MUST OCCUR PRIOR TO EQUIPMENT INSTALLATION/REMOVAL.
2. IF A HAZARDOUS CHEMICAL (MATERIAL) AS DEFINED UNDER 29 CFR 1910.1200 FOR THE INSTALLATION, TESTING, OR OPERATION OF INSTALLED MACHINERY IS REQUIRED, THE CONTRACTOR MUST RECEIVE APPROVAL FROM THE CONTRACTING OFFICER REPRESENTATIVE PRIOR TO ANY OF THAT MATERIAL BEING BROUGHT ON SITE. SUBMIT IN WRITING THE FOLLOWING:
 - 2.1. SAFETY DATA SHEET (SDS), QUANTITY, INTENDED USE, AND OTHER INFORMATION THAT MAY BE REQUIRED. MAINTAIN COPIES OF ALL SDS ON SITE THROUGHOUT THE PROJECT.
 - 2.2. ENSURE CONTRACTOR EMPLOYEES UNDERSTAND HOW TO LOCATE A SDS. COMPLETE FRC EAST FORM 5090/20 AND PROVIDE TO THE CONTRACTING OFFICER REPRESENTATIVE.
3. HAZARDOUS WASTE GENERATED MUST BE STORED, HANDLED AND DISPOSED IN ACCORDANCE WITH FEDERAL HAZARDOUS WASTE LAWS FOUND AT 40 CFR 260-268. CONTACT THE CONTRACTING OFFICER REPRESENTATIVE FOR ADDITIONAL INSTRUCTION.
4. THIS SITE IS WITHIN THE BOUNDARIES OF OPERABLE UNIT 1 (OU1) AND IS WITHIN OR NEAR KNOWN AREAS OF SOIL CONTAMINATION. IF EARTH IS DISTURBED IT SHOULD BE TREATED AS CONTAMINATED. SEE DETAILS WITHIN THE CONTRACT DOCUMENTS FOR ADDITIONAL INFORMATION.
5. FOR CONNECTIONS TO OR MODIFICATION OF EXISTING DRAINS OR DRAINAGE SYSTEMS (E.G. SANITARY SEWER, INDUSTRIAL SEWER, OR STORM SEWER), CONTACT THE CONTRACTING OFFICER REPRESENTATIVE FOR APPROVAL OF THE CONNECTION AND FINAL DISCHARGE LOCATION.
6. CONNECTION TO THE POTABLE WATER SYSTEM MAY BE PERFORMED USING A BACKFLOW PREVENTOR. INSTALLATION OF BACKFLOW PREVENTION ASSEMBLIES MUST BE ON THE LIST OF APPROVED ASSEMBLIES BY THE UNIVERSITY OF SOUTHERN CALIFORNIA FOUNDATION OF CROSS-CONTROL AND HYDRAULIC RESEARCH (USC FCCC HR) AND SHALL BE IN COMPLIANCE WITH THE INTERNATIONAL PLUMBING CODE (IPC) SECTION 608. NOTIFY THE LOCAL CROSS-CONNECTION OPERATOR IN RESPONSIBLE CHARGE (ORC) AUTHORITY VIA THE COR WHEN INSTALLATION IS COMPLETE. THE BACKFLOW PREVENTION ASSEMBLY MUST THEN BE TESTED AND CERTIFIED BY THE LOCAL AUTHORITY.

SPECIAL ENVIRONMENTAL NOTES:

1. THIS PROJECT IS LOCATED WITHIN THE DEFINED LIMITS OF OPERABLE UNIT 1. THIS OPERABLE UNIT HAS A RECORD OF DECISION (ROD) THAT REQUIRES CERTAIN REPORTING ACTIONS BE COMPLETED IN ORDER TO BE IN COMPLIANCE WITH THE ROD. NO DEWATERING ACTIVITIES CAN OCCUR IN THESE AREAS WITHOUT PRIOR WRITTEN APPROVAL. SEE SPECIFICATIONS SECTION 01 14 00 "WORK RESTRICTIONS".
2. SPECIFICATION SECTION 31 00 00 "EARTHWORK" OUTLINES THE REQUIREMENTS FOR HANDLING SUSPECTED CONTAMINATED SOILS AND ALL FILL MATERIALS IMPORTED TO THE BASE. EXCAVATED SOIL MUST BE STOCKPILED AND TESTED.

OCCUPATIONAL SAFETY & HEALTH NOTES:

1. ASBESTOS MATERIAL AND LEAD-BASED PAINT TESTING HAS BEEN PERFORMED ON EXISTING AREAS OF WORK. SEE THE SPECIFICATION FOR ASBESTOS MATERIAL AND LEAD-BASED PAINT REPORT RESULTS. CONTRACTOR IS RESPONSIBLE FOR THE APPROPRIATE STANDARD OF CARE REGARDING THE HANDLING OF ANY MATERIALS FOUND TO BE HAZARDOUS AND THE PROTECTION OF PERSONS FROM EXPOSURE.
2. COMPLY WITH SAFETY AND OCCUPATIONAL CODE REQUIREMENTS INCLUDING; ANSI A10.32 PERSONAL FALL PROTECTION, EM 385-1-1 SAFETY AND HEALTH REQUIREMENTS, 29 CFR 1910 OCCUPATIONAL SAFETY AND HEALTH STANDARDS, 29 CFR 1926 SAFETY AND HEALTH REGULATIONS FOR CONSTRUCTION, 29 CFR 1926.500 FALL PROTECTION P-307 MANAGEMENT OF WEIGHT HANDLING EQUIPMENT, NFPA 70 NATIONAL ELECTRIC CODE, AND NFPA 70E STANDARDS FOR ELECTRICAL SAFETY REQUIREMENTS FOR EMPLOYEE WORKPLACES.

SECURITY NOTES:

1. WORK IS WITHIN THE FRC EAST SECURITY ZONE. OBTAIN SECURITY BADGES IN ACCORDANCE WITH CONTRACT DOCUMENTS.

SYMBOL LEGEND

DRAWING

SCALE: FULL SCALE

DRAWING TAG

XX

DETAIL / ENLARGEMENT

SCALE: FULL SCALE

DETAIL TAG

12'-0"

①

ROOM
402

Y

DIMENSIONS SHOWN THUSLY ARE TO THE
FACE OF THE ELEMENT BEING DIMENSIONED
(WALL, FRAME, ETC)

NUMBER TAG

KEYNOTE TAG

ROOM DESIGNATION

COLUMN LINE DESIGNATION

ELEVATION MARK

SECTION MARK

NORTH ARROW

ENLARGED PLAN / DETAIL MARK

BREAK

[illegible]

1

2

3

4

5

GENERAL

1.

NO PIPES OR DUCTS WILL BE PLACED IN STRUCTURAL MEMBERS UNLESS SPECIFICALLY DETAILED OR APPROVED BY THE ENGINEER & CONTRACTING OFFICER.

2.

REFER TO ARCHITECTURAL DRAWINGS FOR THE FOLLOWING:

2.1.

SIZE AND LOCATION OF ALL DOOR AND WINDOW OPENINGS, UNLESS OTHERWISE NOTED.

2.2.

SIZE AND LOCATION OF INTERIOR AND EXTERIOR NON-BEARING PARTITIONS.

2.3.

SIZE AND LOCATION OF CURBS, FLOOR DRAINS, SLOPES, DEPRESSED AREAS, CHANGES IN LEVEL, RAMPS, CHAMFERS, GROOVES, INSERTS, ETC., EXCEPT AS SHOWN.

2.4.

SIZE AND LOCATION OF FLOOR AND ROOF OPENINGS, EXCEPT AS SHOWN.

2.5.

FLOOR AND ROOF FINISHES.

2.6.

STAIR FRAMING AND DETAILS, EXCEPT AS SHOWN.

2.7.

DIMENSIONS NOT SHOWN ON STRUCTURAL DRAWINGS.

3.

REFER TO MECHANICAL, PLUMBING, AND ELECTRICAL DRAWINGS FOR THE FOLLOWING:

3.1.

PIPE RUNS, SLEEVES, HANGERS, TRENCHES, SLAB OPENINGS, ETC., EXCEPT AS SHOWN OR NOTED.

3.2.

ELECTRICAL CONDUITS, BOXES, OUTLETS. CONCRETE INSERTS FOR ELECTRICAL, MECHANICAL AND PLUMBING FIXTURES.

3.3.

SIZE AND LOCATION OF MACHINE AND EQUIPMENT BASES, ANCHOR, BOLTS, ETC.

4.

INVESTIGATE THE SITE DURING CLEARING AND EXCAVATION FOR UNSUITABLE CONDITIONS, UNCONSOLIDATED AND UNDOCUMENTED FILLS, BURIED STRUCTURES, UTILITIES, ETC., AND IMMEDIATELY NOTIFY THE ARCHITECT/ENGINEER & CONTRACTING OFFICER OF ANY SITE CONDITIONS NOT REFLECTED ON THE DRAWINGS OR DIFFERENT FROM MAXIMUM OR MINIMUM DIMENSIONS INDICATED, INCLUDING CONFLICT IN GRADES, ADVERSE SOIL CONDITIONS, GROUNDWATER PRESENT, DEEPENED FOOTINGS, UNCOVERED AND UNEXPECTED UTILITY LINES, ETC.

5.

CONSTRUCTION MATERIALS, IF PLACED ON STRUCTURAL MEMBERS, WILL BE SPREAD OUT SUCH THAT THE LOADING DOES NOT EXCEED THE DESIGN LIVE LOADS. PROVIDE SHORING AND BRACING WHERE CONSTRUCTION LOADING EXCEEDS THE DESIGN STRENGTH OF THE STRUCTURAL MEMBERS OR THE STRUCTURAL STRENGTH HAS NOT BEEN ATTAINED OR THE STRUCTURE IS NOT COMPLETE.

6.

DETERMINE THE LOCATION OF UTILITY SERVICES IN AREAS TO BE EXCAVATED BEFORE BEGINNING EXCAVATION. EXERCISE EXTREME CAUTION IN EXCAVATING AND TRENCHING. DAMAGE CAUSED AS A RESULT OF FAILING TO EXACTLY LOCATE AND PRESERVE EXISTING UNDERGROUND UTILITIES IS THE RESPONSIBILITY OF THE CONTRACTOR.

CONCRETE

1.

CEMENT WILL CONFORM TO ASTM C150, TYPE II / V

2.

AGGREGATES FOR NORMAL WEIGHT CONCRETE WILL CONFORM TO ASTM C33, 1 1/2" MAXIMUM SIZE.

3.

ADMIXTURES MAY NOT BE USED WITHOUT PRIOR APPROVAL OF THE ENGINEER & CONTRACTING OFFICER. ADMIXTURES USED TO INCREASE THE WORKABILITY OF THE CONCRETE MUST NOT REDUCE THE STRENGTH OF CONCRETE. FLY ASH (POZZOLAN) IF PERMITTED BY SPECIFICATIONS WILL NOT EXCEED 25%.

4.

READY-MIX CONCRETE WILL BE MIXED AND DELIVERED IN ACCORDANCE WITH ASTM C94.

5.

ALL REINFORCING BARS AND INSERTS WILL BE INSPECTED BY CM IN PLACE PRIOR TO PLACING CONCRETE.

6.

PROVIDE JOINTS IN SLAB ON GRADE AS SHOWN. LOCATION OF JOINTS NOT SPECIFICALLY INDICATED WILL BE REVIEWED BY THE STRUCTURAL ENGINEER & CONTRACTING OFFICER PRIOR TO PLACING REINFORCING STEEL.

7.

MINIMUM CONCRETE COMPRESSIVE STRENGTHS AT 28 DAYS.

	MIN. f'c
SLAB ON GRADE	4,000 psi
OTHER CONCRETE	4,500 psi

REINFORCING STEEL

1.

DETAILING, FABRICATION AND ERECTION OF REINFORCING BARS WILL BE IN ACCORDANCE WITH ACI MANUAL OF STANDARD PRACTICE FOR DETAILING REINFORCED CONCRETE STRUCTURES, ACI 315-LATEST ADOPTED EDITION.

2.

ALL REINFORCING WILL BE ADEQUATELY SUPPORTED TO PREVENT DISPLACEMENT BY CONCRETE PLACEMENT OR WORKERS.

3.

ALL REINFORCING BARS EXCEPT BARS TO BE WELDED WILL CONFORM TO THE "STANDARD SPECIFICATION FOR DEFORMED BILLET STEEL BARS FOR CONCRETE REINFORCEMENT", ASTM A615 GRADE 60. BARS TO BE WELDED WILL CONFORM TO ASTM A706.

4.

WELDING OF REINFORCING BARS TO BE IN ACCORDANCE WITH "STRUCTURAL WELDING CODE-REINFORCING STEEL", AWS D1.4. REINFORCING STEEL TO BE WELDED WILL HAVE A MAXIMUM CARBON EQUIVALENT (CE) OF 0.75. SPECIAL INSPECTION IS REQUIRED. TESTING IS REQUIRED FOR ALL WELDS THICKER THAN 5/16".

5.

WHERE CONTINUOUS BARS ARE CALLED OUT IN FOOTINGS, SPLICES MAY BE USED. WHERE BARS ARE SHOWN SPLICED, THEY MAY RUN CONTINUOUS AT CONTRACTOR'S OPTION.

6.

ALL REINFORCING BAR BENDS WILL BE MADE COLD.

7.

UNLESS OTHERWISE SHOWN, WALL VERTICAL REINFORCING WILL BE POSITIONED AT THE CENTER OF THE WALL.

8.

DOWELS BETWEEN FOOTINGS AND WALLS WILL BE THE SAME GRADE, SIZE, AND SPACING AS VERTICAL REINFORCING.

9.

SHOP DRAWINGS FOR SIZE AND LAYOUT OF REINFORCING REBARS ARE REQUIRED WHEN NOTED IN THE LIST OF REQUIRED SHOP DRAWINGS.

ALL REINFORCING BARS WILL BE PROVIDED WITH THE FOLLOWING CONCRETE COVER:

CONCRETE CAST AGAINST AND PERMANENTLY EXPOSED TO EARTH:..... 3"

CONCRETE EXPOSED TO EARTH OR WEATHER:

NO.6 THROUGH NO.18 BAR 2"

NO.5 BAR, W31 OR D31 WIRE, AND SMALLER 1 1/2"

CONCRETE NOT EXPOSED TO WEATHER AND NOT IN CONTACT WITH GROUND:

NO.14 AND NO.18 BAR 1 1/2"

NO.11 BAR AND SMALLER 3/4"

EPOXY ANCHORS

1.

ADHESIVE ANCHOR INSTALLATIONS WILL COMPLY WITH MANUFACTURER'S WRITTEN INSTRUCTIONS AND SPECIFICATIONS, INCLUDING ICC-ES REPORTS.

2.

CHEMICAL ANCHOR SYSTEMS:

2.1.

CONCRETE: USE ONLY ADHESIVE ANCHOR SYSTEMS THAT HAVE BEEN ISSUED AN ICC-ES REPORT IN ACCORDANCE WITH PROVISIONS OF OF ICC-ES AC308. ANCHOR SYSTEM SHOULD BE APPROVED FOR USE IN CRACKED CONCRETE AND SEISMIC DESIGN CATEGORIES A-F PER SECTION 2.0 OF THE ICC-ES EVALUATION SERVICES REPORT. ANCHOR SYSTEM WILL BE INSTALLED PER REQUIREMENTS OF THE ICC-ES EVALUATION SERVICES REPORT FOR SPECIFIC ANCHOR, AND AS REQUIRED BY THE MANUFACTURER.

2.2.

GROUT-FILLED MASONRY UNITS: USE ONLY ADHESIVE ANCHOR SYSTEMS THAT HAVE BEEN ISSUED AN ICC-ES REPORT IN ACCORDANCE WITH PROVISIONS OF ISS-ES AC58. ANCHOR SYSTEMS WILL BE INSTALLED PER REQUIREMENTS OF ICC-ES EVALUATION SERVICES REPORT FOR THE SPECIFIC ANCHOR, AND AS REQUIRED BY THE MANUFACTURER.

3.

ANCHOR RODS: RODS WILL BE ASTM A36 THREADED RODS WITH ASTM A563 GRADE A NUTS AND ANSI B18.22.1 TYPE A WASHERS, UNLESS OTHERWISE NOTED. ANCHORS DESIGNATED AS ASTM A193 GRADE B7 THREADED RODS WILL USE ASTM A563 GRADE DH HEAVY HEX NUTS AND ASTM F436 WASHERS.

4.

REINFORCEMENT BARS: ASTM A615 GRADE 60 STEEL.

5.

REMOVE GREASE, OIL, RUST AND ANY OTHER LAITANCE FROM RODS AND DOWELS PRIOR TO INSTALLATION.

STRUCTURAL STEEL

1.

ALL STRUCTURAL STEEL SHALL BE NEW, CLEAN AND STRAIGHT. STEEL WIDE FLANGE SHAPES SHALL CONFORM TO ASTM A-992, GRADE 50. ALL OTHER STEEL SHAPES MUST CONFORM TO ASTM 36 UNLESS OTHERWISE NOTED.

2.

WELDING MUST COMPLY WITH THE "STRUCTURAL WELDING CODE" PREPARED BY THE A.W.S., MINIMUM WELD SIZE SHALL BE 3/16" FILLET WELD, ELECTRODES SHALL BE E70XX.

3.

UNLESS OTHERWISE NOTED HIGH STRENGTH BOLTS SHALL COMPLY WITH ASTM A-325 AND ANCHOR BOLTS SHALL COMPLY WITH ASTM A-307 OR ASTM A-36.

4.

ALL STRUCTURAL STEEL, GRATING, STAIRS, BOLTS, NUTS AND WASHERS SHALL BE GALVANIZED. HOT-DIP GALVANIZE AFTER FABRICATION WHERE PRACTICAL. GALVANIZING MUST CONFORM TO ASTM A123 OR ASTM A153 AS APPLICABLE. APPLY ASTM A-780 ZINC RICH PAINT TO DAMAGED GALVANIZED COATINGS AND WELDS WHERE APPLICABLE.

DESIGN CRITERIA

1.

UFC 3-310-01, STRUCTURAL ENGINEERING

DESIGN LOADS

DESIGN LIVE LOADS USED IN THE DESIGN OF THIS STRUCTURE ARE AS FOLLOWS:

1.

UNIFORM LIVE LOADS:

ROOF: 20 PSF

PLATFORMS AND FLOOR FRAMING: 60 PSF

2.

WIND LOADS:

WIND VELOCITY (3 SEC GUST) 139 MPH

IMPORTANCE FACTOR I = 1.0, BUILDING CATEGORY II

WIND EXPOSURE CATEGORY: C

INTERNAL PRESSURE COEFFICIENT: +/- 0.55 (PARTIALLY ENCLOSED)

COMPONENTS AND CLADDING: WIND PRESSURE TO BE USED FOR DESIGN OF EXTERIOR COMPONENTS AND CLADDING MATERIALS NOT SPECIFICALLY DESIGNED ON THESE DRAWINGS MUST BE 50 PSF.

SNOW LOADS:

GROUND SNOW LOAD: 10 PSF

SEISMIC LOADS:

IMPORTANCE FACTOR: 1.0

0-2 SECOND SPECTRAL RESPONSE, S_s = 16%g

1-0 SECOND SPECTRAL RESPONSE, S_i = 7%g

SOIL SIDE CLASS = D (ASSUMED)

LEGEND AND ABBREVIATIONS

EXISTING

DEMOLITION/CONSTRUCTION WORK

DEMOLITION/CONSTRUCTION WORK

ARCHITECTURAL

APPROXIMATELY

CENTERLINE

CONSTRUCTION MANAGER

DIAM

DIMENSIONS

EDGE OF CONCRETE

FOOT

MAXIMUM

MINIMUM

MILES PER HOUR

ON CENTER

POUNDS PER CUBIC FOOT

POUNDS PER SQUARE FOOT

POUNDS PER SQUARE INCH

SECOND

SLAB-ON-GRADE

SQUARE FOOT

TYPICAL

ARCH

APPROX.

€

CM

DIAMETER

DIMS

E.O.C.

FT

MAX

MIN

MPH

O.C.

PCF

PSF

PSI

SEC

S.O.G

SQ FT

TYP

DEPARTMENT OF THE NAVY

NAVAL FACILITIES ENGINEERING COMMAND

FLEET READINESS CENTER EAST

FACILITIES INFRASTRUCTURE PLANNING & ENGINEERING BRANCH

PROJECT STATION

U.S. MARINE CORPS AIR STATION, CHERRY POINT, NC

FACILITY PREPARATION AND PROVIDE ULTRASONIC CLEANING SYSTEM

STRUCTURAL NOTES

SCALE: AS NOTED

CONSTR. CONTR. NO. #####-##-###

NAVFAC DRAWING NO. 12892405

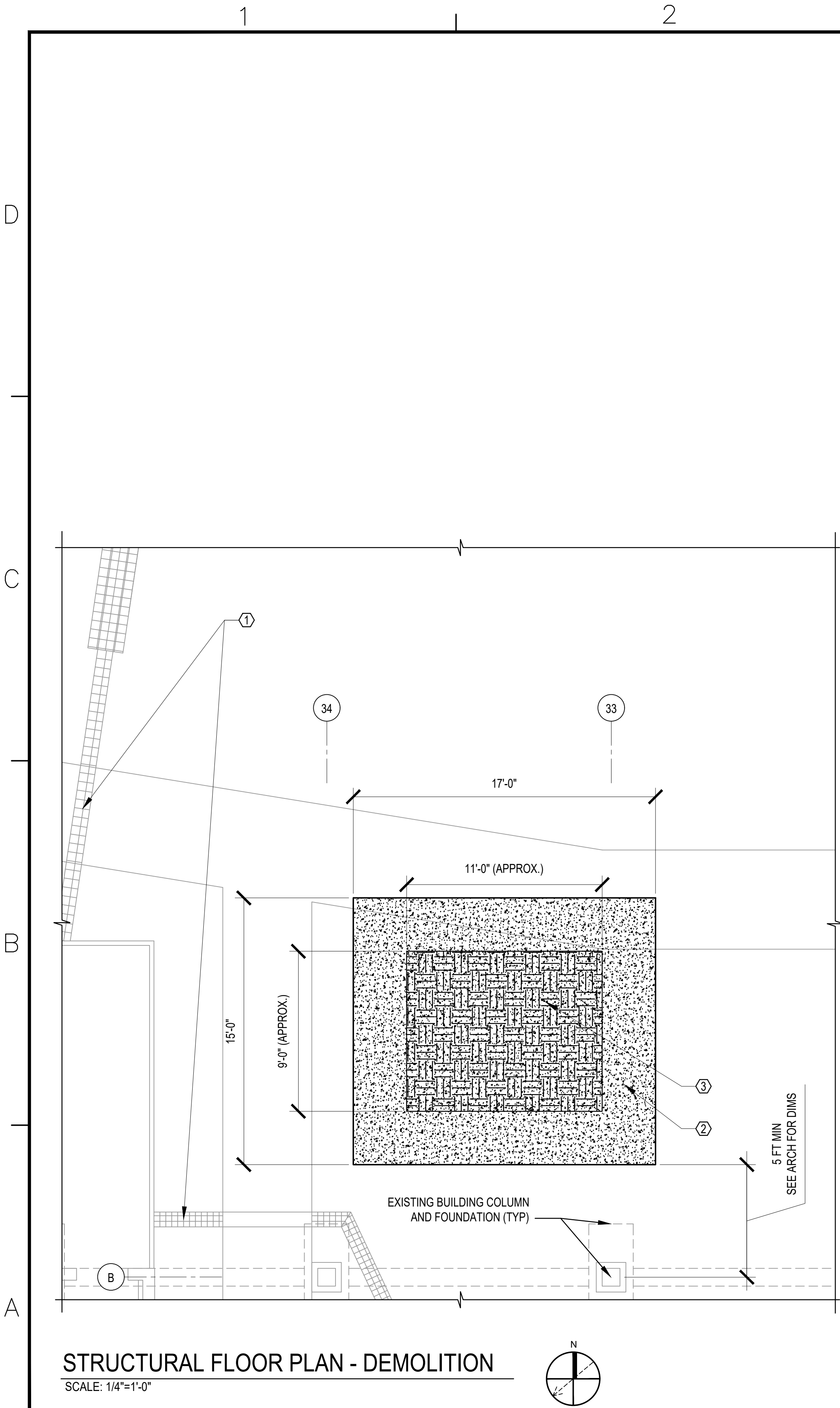
CP FEAD. NO.: 7357358

FRCE NO.: XZ82190

SHEET 3 OF 19

S-001

DRAWFORM REVISION: 24 JULY 2019



GENERAL SHEET NOTES

- FOR GENERAL NOTES REFER TO SHEET G-002 AND S-001.
- REFER TO G-002 FOR SPECIAL ENVIRONMENTAL NOTES.
- THE CONTRACTOR IS RESPONSIBLE FOR LOCATING UNDERGROUND UTILITIES IN THE AREA OF EXCAVATION. IF AN UNEXPECTED UNDERGROUND UTILITY IS FOUND, PROTECT FROM DAMAGE AND CONTACT THE CONTRACTING OFFICER REPRESENTATIVE.
- DO NOT UNDERMINE EXISTING BUILDING FOUNDATIONS.
- CAPACITIES OF EXISTING FLOOR GRATES ARE UNKNOWN. PROTECT FROM DAMAGE.
- DIMENSIONS SHOWN ON PLAN AS APPROXIMATE MUST BE COORDINATED WITH PRECAST CONCRETE VAULT INSTALLATION. THE CONTRACTOR MUST DETERMINE FINAL REQUIRED DIMENSIONS AND SUBMIT TO THE COR FOR APPROVAL.

KEYNOTES

APPLIES TO THIS SHEET ONLY

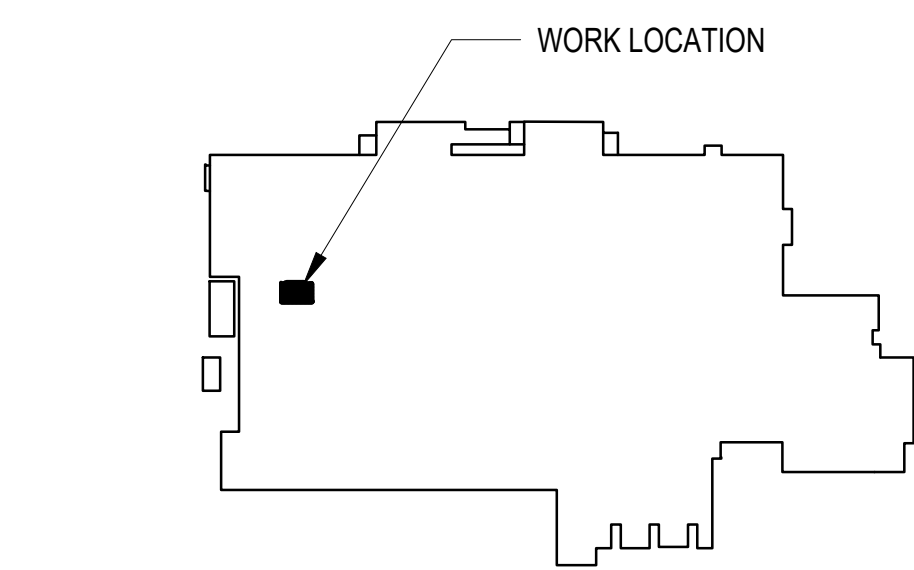
- PROVIDE TEMPORARY SUPPLEMENTAL MATERIAL ATOP EXISTING FLOOR GRATES WHEN CROSSING WITH HEAVY LOADS.
- REMOVE EXISTING 6 INCH THICK CONCRETE SLAB-ON-GRADE AND EXISTING AGGREGATE BASE CORSE IN THIS AREA TO ACCOMMODATE PRECAST CONCRETE VAULT.
- EXCAVATE EXISTING SUBGRADE IN THIS AREA TO ACCOMMODATE PRECAST CONCRETE VAULT.



7/18/2024
FINAL
DIVISION APPROVED
WHEN SIGNED

APPROVED	
FOR COMMANDER NAVFAC	
BES	A. WILLIAMS
DRW	A. WILLIAMS
CHK	J. PADRO
REVIEWED	A. PAU
APPROVED	E. LEWIS
FAC PLAN PROJ NO: PE-22034	
SHOP: 93112	CAGE CODE: 96916

DEPARTMENT OF THE NAVY
FLEET READINESS CENTER EAST
FACILITIES INFRASTRUCTURE PLANNING & ENGINEERING BRANCH
PROJECT STATION
NAVAL FACILITIES ENGINEERING COMMAND
MCAS CHERRY POINT, NC
U.S. MARINE CORPS AIR STATION, CHERRY POINT, NC
FACILITY PREPARATION AND
PROVIDE ULTRASONIC CLEANING SYSTEM
STRUCTURAL FLOOR PLAN - DEMOLITION



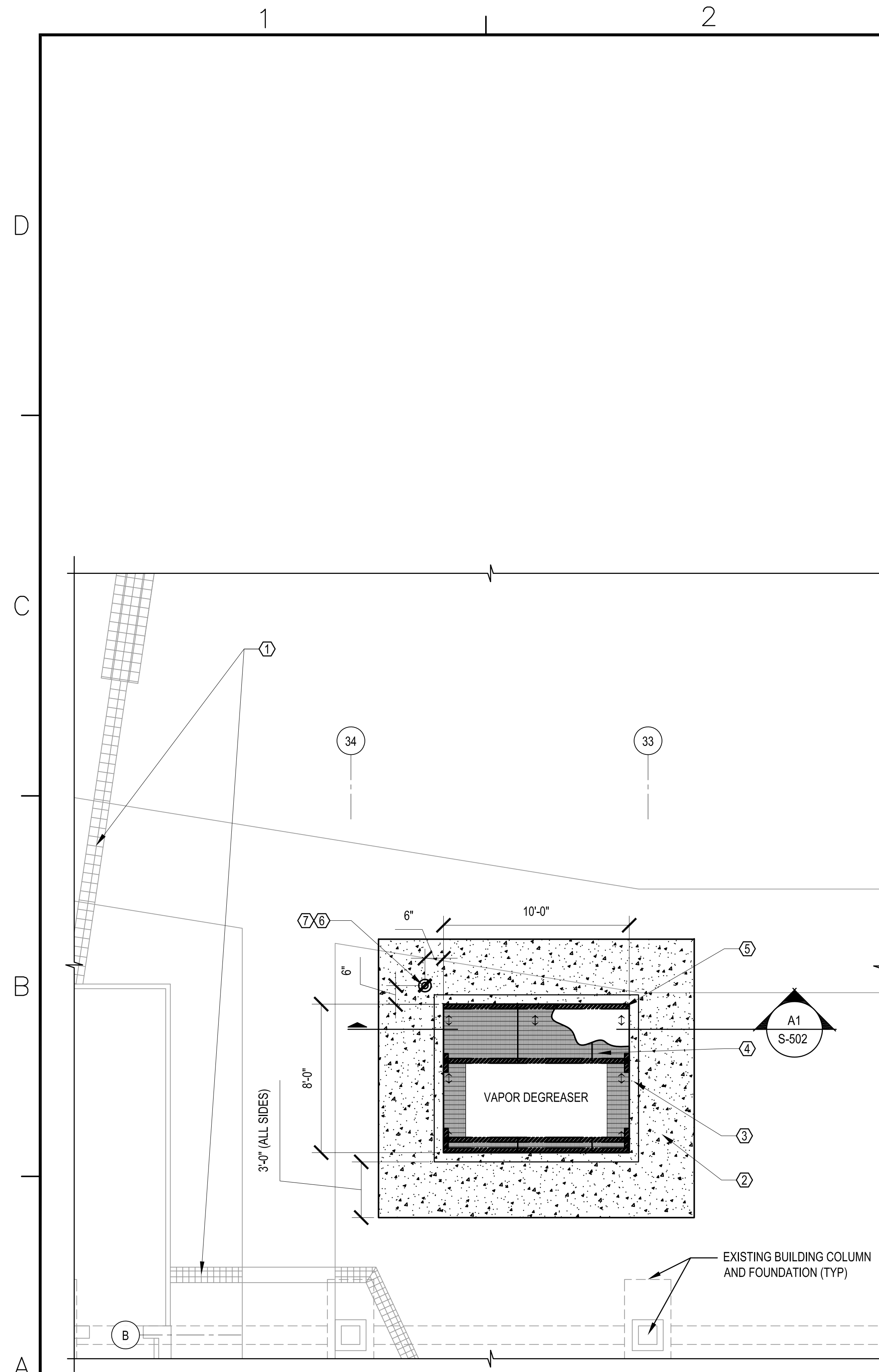
B133 WORK LOCATION
SCALE: NOT TO SCALE



GRAPHIC SCALE

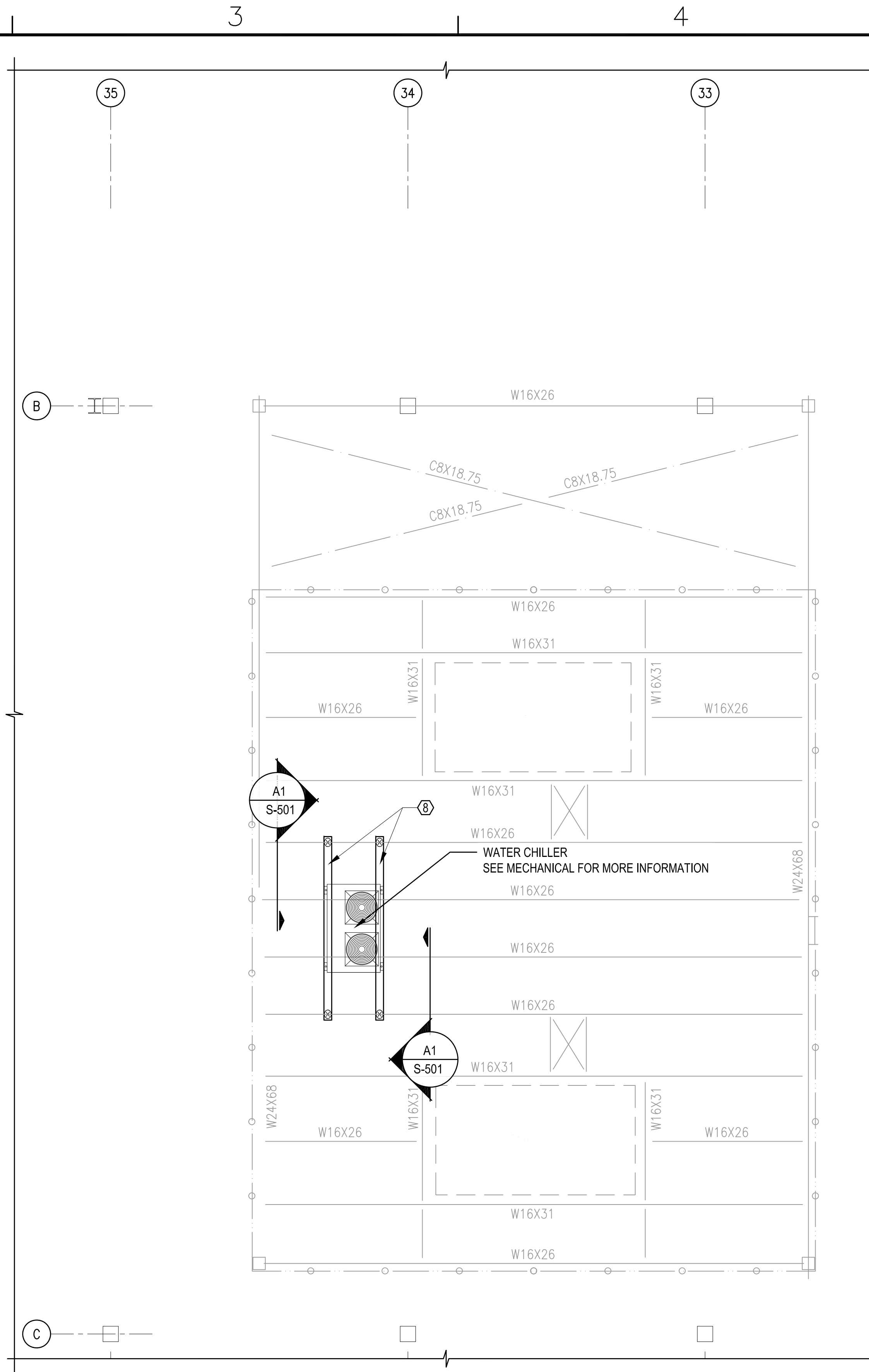
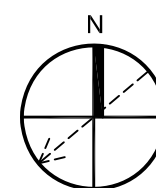


SCALE:	AS NOTED
CONSTR. CONTR. NO.	####-##-##-####
NAVFAC DRAWING NO.	12892406
CP FEAD NO:	7357358
TRCE NO:	XZ62190
SHEET	4 OF 19
SD101	



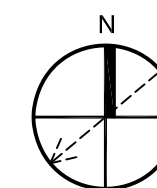
STRUCTURAL FLOOR PLAN - CONSTRUCTION

SCALE: 1/4"=1'-0"



STRUCTURAL ROOF PLAN - CONSTRUCTION

SCALE: 1/4"=1'-0"



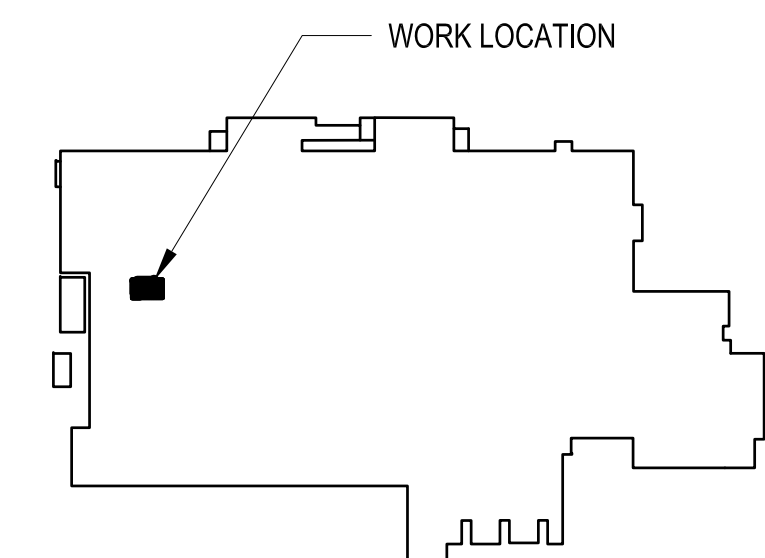
GENERAL SHEET NOTES

1. FOR GENERAL NOTES REFER TO SHEET G-002 AND S-001.
2. REFER TO G-002 FOR SPECIAL ENVIRONMENTAL NOTES.
3. RELEVANT DIMENSIONS OF EXISTING PLATFORM STRUCTURE MUST BE VERIFIED PRIOR TO FABRICATION OF STEEL BEAMS FOR EQUIPMENT SUPPORT.
4. DO NOT UNDERMINE EXISTING BUILDING FOUNDATIONS.
5. CAPACITIES OF EXISTING FLOOR GRATES ARE UNKNOWN. PROTECT FROM DAMAGE.
6. GRATING MAIN BARS SPANNING AS INDICATED THUS (⌄) ON PLAN.
7. STEEL EXPOSED TO WEATHER MUST BE GALVANIZED.
8. APPLY ASTM A-780 ZINC RICH PAINT TO DAMAGED GALVANIZED COATINGS AND WELDS.
9. CHILLER WEIGHT MUST NOT EXCEED 2,500LBS.

KEYNOTES

APPLIES TO THIS SHEET ONLY

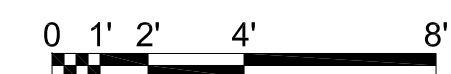
- ① PROVIDE TEMPORARY SUPPLEMENTAL MATERIAL ATOP EXISTING FLOOR GRATES WHEN CROSSING WITH HEAVY LOADS.
- ② PATCH CONCRETE SLAB-ON-GRADE. PROVIDE DOWEL CONSTRUCTION JOINTS AND ISOLATION JOINTS AS REQUIRED.
- ③ PROVIDE 8 FT X 10 FT X 3FT DEEP (INTERIOR DIMENSIONS) PRECAST CONCRETE VAULT .
- ④ PROVIDE 8FT LONG I-BAR 4015 PULTRUDED GRATING, VINYL ESTER FIRE RETARDANT (VEFR). ORIENT GRATING AS INDICATED. PROVIDE GRATING ACCESSORIES INCLUDING CLIPS AND FASTENERS. ATTACH GRATING TO SUPPORT FRAMING PER MANUFACTURERS RECOMMENDATIONS.
- ⑤ PROVIDE STEEL SUPPORT FRAMING FOR GRATING. SEE ENLARGED PLAN B1/S-501 FOR SUPPORT FRAMING.
- ⑥ PROVIDE PERMANENT MOUNT DELINEATOR POST WITH GRABBER/ LOOPER TOP. POST MUST BE POLYETHYLENE, COLOR ORANGE, 4" DIAMETER, 42" TOTAL HEIGHT WITH 8" DIAMETER BASE. MOUNT BASE TO CONCRETE PER MANUFACTURER'S RECOMMENDATIONS. LOCATE POST 6" OFF VAULT CORNER AS SHOWN.
- ⑦ ATTACH CAPACITY SIGN TO TOP LOOP OF POST. SIGN FURNISHED BY GOVERNMENT.
- ⑧ PROVIDE TWO (2) W6X20 STEEL BEAMS ATOP EXISTING STEEL GRATE DECKING. W6X20 BEAM MUST SPAN FOUR (4) EXISTING BEAMS AS SHOWN. FASTEN EACH BEAM TO EXISTING PLATFORMS BEAMS AT LOCATIONS INDICATED THUS (⊗). ATTACH CHILLER TO W6X20 BEAMS PER CHILLER MANUFACTURER'S RECOMMENDATIONS.



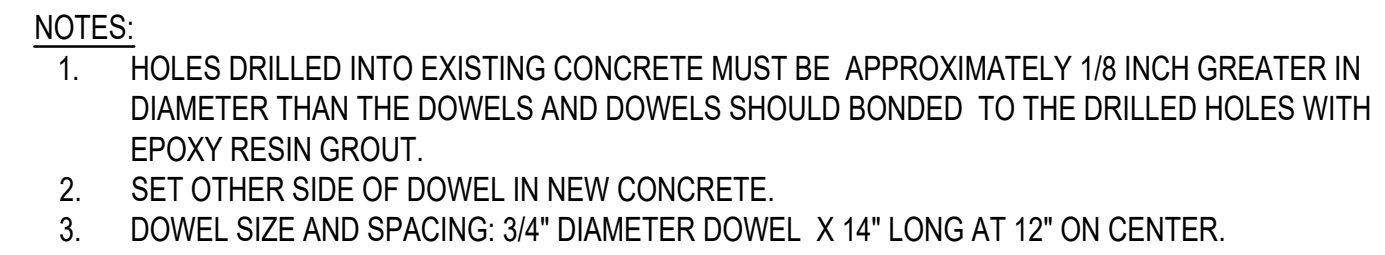
B133 WORK LOCATION

SCALE: NOT TO SCALE

GRAPHIC SCALE

$$1/4" = 1'-0"$$


FLEET READINESS CENTER		EAST		FRC		EFFICIENCY • AGILITY • RESILIENCE	
7/18/2024 FINAL DIVISION APPROVED WHEN SIGNED							
APPROVED FOR COMMANDER NAVFAC DES A. WILLIAMS DRW A. WILLIAMS CHK J. PADRO REVIEWED A. PAU APPROVED E. LEWIS FAC PLAN PROJ NO: PE-22034 SHOP: 93112 CASE CODE: 96916							
DEPARTMENT OF THE NAVY FLEET READINESS CENTER EAST NAVAL FACILITIES ENGINEERING COMMAND FACILITIES INFRASTRUCTURE PLANNING & ENGINEERING BRANCH		LOCUS CHERRY POINT, NC PROJECT STATION U.S. MARINE CORPS AIR STATION, CHERRY POINT, NC		FACILITY PREPARATION AND PROVIDE ULTRASONIC CLEANING SYSTEM			
SCALE: AS NOTED CONSTR. CONTR. NO. NAVFAC DRAWING NO. 12892407 GP FEAD WO: 7357358 FRCE WO: XZ62190 SHEET 5 OF 19		STRUCTURAL FLOOR & ROOF PLAN - CONSTRUCTION					
S-101							



6X6 WWF 10/10

CONCRETE SLAB-ON-GRADE PATCH

COMPACTED AGGREGATE BASE COURSE

#57 GRANITE STONE

2"

6" MIN.

MATCH EXISTING ADJACENT S.O.G. THICKNESS IF 6" OR GREATER

15 MIL MINIMUM THICKNESS VAPOR BARRIER CONTINUOUS UNDER S.O.G. AND VAULT

COMPACTED FILL SOIL OR AGGREGATE

POURED JOINT SEALANT
SEALANT RESERVOIR:
WIDTH = 1/2"
DEPTH = 1.0 TO 1.5 TIMES THE WIDTH
TOP OF SEALANT WILL BE 1/8" BELOW TOP OF PAVEMENT

CONCRETE VAULT

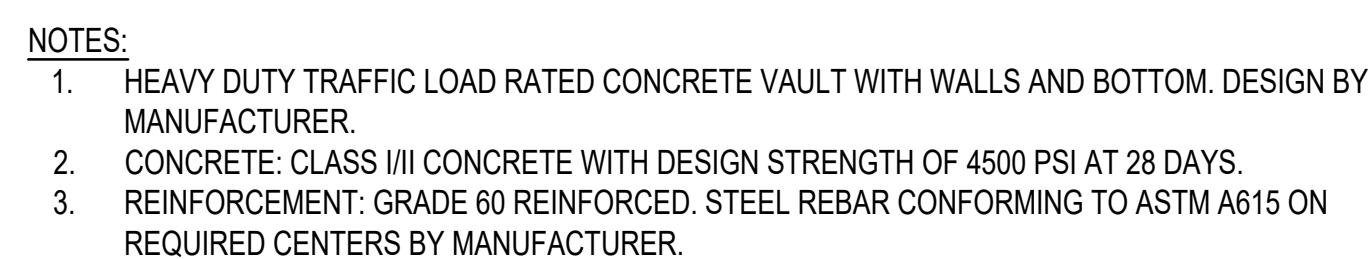
CONCRETE S.O.G

BONDBREAKER
OR BACKER ROD

1/2 INCH THICK
JOINT MATERIAL

The diagram is a cross-sectional view of a concrete vault joint. It shows two concrete sections: a 'CONCRETE VAULT' on the left and a 'CONCRETE S.O.G' (Subgrade or Support) on the right. A vertical joint separates them. At the top of the joint, there is a 'POURED JOINT SEALANT SEALANT RESERVOIR' with a width of 1/2 inch and a depth of 1.0 to 1.5 times the width. The top of the sealant is 1/8 inch below the top of the pavement. A 'BONDBREAKER OR BACKER ROD' is shown as a vertical line within the sealant reservoir. Below the sealant, there is a '1/2 INCH THICK JOINT MATERIAL' layer. The bottom of the vault and S.O.G. are shown with a break symbol, indicating they continue below the shown section.

3/4" = 1'-0"

A horizontal graphic scale bar with alternating black and white segments. It is marked with '0' at the left end, and '1'', '2'', and '3'' at subsequent intervals. The total length represents 3 feet.

SCALE: 3/4"=1'-0'

DRAWFORM REVISION: 24 JULY 2019

1. REVIEW DOCUMENTS, VERIFY ALL DIMENSIONS AND FIELD CONDITIONS AND CONFIRM THAT WORK IS CONSTRUCTIBLE AS SHOWN. REPORT ANY CONFLICTS OR OMISSIONS TO THE UNITED STATES GOVERNMENT (USG) DESIGNATED CONTRACTING OFFICER'S REPRESENTATIVE (COR) FOR CLARIFICATION PRIOR TO PERFORMING ANY WORK IN QUESTION.
2. COORDINATE ALL WORK WITH COR, INCLUDING SCHEDULING TIME AND LOCATIONS FOR DELIVERIES, BUILDING ACCESS, USE OF BUILDING SERVICES AND FACILITIES. MINIMIZE DISTURBANCE OF BUILDING FUNCTIONS AND/OR OCCUPANTS.
3. PROTECT AREA OF WORK AND ALL ADJACENT AREA FROM DAMAGE.
4. PROTECT ALL MATERIAL UNLESS OTHERWISE NOTED FOR DEMOLITION.
5. REPAIR AND RESEAL CONCRETE WHERE EXISTING PAINT, WALLS, AND COMPONENTS HAVE BEEN REMOVED OR RECONFIGURED UNLESS OTHERWISE NOTED.
6. VERIFY ALL SURROUNDINGS AND CONCEALED ITEMS PRIOR TO DEMOLITION. ALL ITEMS DAMAGED DURING DEMOLITION THAT ARE TO REMAIN WILL BE THE RESPONSIBILITY OF THE CONTRACTOR TO REPLACE TO ORIGINAL STATE. IF DEMOLITION IS PERFORMED IN EXCESS OF THAT REQUIRED, RESTORE AFFECTED AREAS AT NO COST TO THE USG.
7. MAINTAIN ALL EXITS, EXIT LIGHTING, PATHS OF EGRESS, FIRE PROTECTIVE DEVICES AND ALARMS IN CONFORMANCE WITH CODES AND ORDINANCES.
8. MAINTAIN SECURE AND LOCKABLE WORK AREAS BOTH DURING AND AFTER WORK HOURS. COORDINATE WITH COR TO ENSURE SECURITY.
9. COMPLY WITH APPLICABLE LOCAL, STATE AND FEDERAL CODES AND REGULATIONS PERTAINING TO SAFETY OF PERSONS, PROPERTY AND ENVIRONMENTAL PROTECTION.
10. DIMENSIONS OF EXISTING CONDITIONS ARE ESTABLISHED BASED ON RECORD DRAWINGS AND SITE SURVEY, DO NOT SCALE DRAWINGS.
11. PROVIDE AND MAINTAIN BARRICADES, LIGHTING, AND GUARDRAILS AS REQUIRED BY APPLICABLE CODES AND REGULATIONS TO PROTECT OCCUPANTS OF BUILDING AND WORKERS.
12. REMOVE FROM SITE DAILY AND LEGALLY DISPOSE OF REFUSE, DEBRIS, RUBBISH AND OTHER MATERIALS RESULTING FROM DEMOLITION OPERATIONS.
13. DEMOLITION WORK SHALL NOT INTERRUPT POWER, TELECOM, ALARM SYSTEMS OR OTHER UTILITIES SERVING AREAS OUTSIDE THE LIMITS OF WORK UNLESS COORDINATED PER SPECIFICATIONS.
14. PROTECT EXISTING EQUIPMENT AND FACILITIES. PROVIDE TEMPORARY CONSTRUCTION BARRIERS AND DUST PARTITIONS TO PREVENT THE SPREAD OF PARTICULATES DURING CONSTRUCTION. PROVIDE TEMPORARY SIGNAGE INDICATING THE CONSTRUCTION ZONE AND WARNING SIGN INDICATING THAT ONLY AUTHORIZED CONSTRUCTION PERSONNEL ARE AUTHORIZED TO ENTER THE CONSTRUCTION AREA.
15. PROVIDE TEMPORARY PROTECTION AGAINST MOISTURE, DUST, DEBRIS AND NOISE IN OCCUPIED SPACES ADJACENT TO DEMOLITION AND CONSTRUCTION AREAS.
16. WHERE PIPES, CONDUITS, DUCTS, OR OTHER UTILITIES ARE REMOVED AND NOT REPLACED, PROVIDE MATERIALS TO FILL THE OPENING(S) THAT ARE SIMILAR TO ADJACENT CONSTRUCTION.
17. FIELD VERIFY DIMENSIONS AND EXISTING CONDITIONS BEFORE CONSTRUCTION
18. WORK ASSOCIATED WITH THIS PACKAGE (I.E. - DELIVERY DATES, INSTALLATION, RELOCATIONS, ETC) MUST BE COORDINATED THROUGH THE PROJECT MANAGER, ALIKA PAU, 252.464.8828.

	VIEW NUMBER	INTERIOR ELEVATION
	WALL TYPE TAG	
	DOOR TAG	
	LEVEL	ELEVATION
	2' x 2' ACOUSTICAL TILE CEILING SYSTEM AT 8'-0" AFF	
	2' x 4' LAY-IN LIGHT FIXTURE	
	MAKE UP AIR FIXTURE	
	EXHAUST AIR FIXTURE	

ABA	ARCHITECTURAL BARRIERS ACT
ACG	ACOUSTICAL CEILING GRID
ACP	ACOUSTICAL CEILING PANELS
ACT	ACOUSTICAL CEILING TILE
AFF	ABOVE FINISHED FLOOR
CAC	COMMON ACCESS CARD
CG	CORNER GUARD
COR	CONTRACTING OFFICER REPRESENTATIVE
CMU	CONCRETE MASONRY UNIT
EC	EXISTING CONCRETE
ELEV A.G.	ELEVATION ABOVE GROUND
ELEV A.S.	ELEVATION ABOVE EXISTING SLAB
EP	EPOXY
ESD	ELECTROSTATIC DISSIPATIVE RUBBER FLOORING
FE	FIRE EXTINGUISHER
FFE	FURNITURE, FIXTURES & EQUIPMENT
GB	GYPSUM BOARD
GBG	GYPSUM BOARD GRID
HM	HOLLOW METAL
HWS#	HARDWARE SET #
MAX	MAXIMUM
MIN	MINIMUM
MTL	METAL
OC	ON CENTER
PNT	PAINT
RB	RUBBER BASE
REF	REFERENCE
R-VALUE	RESISTANCE VALUE
SPEC	SPECIFICATIONS
TELE	TELEPHONE
TS	TRANSITION STRIP
TYP	TYPICAL
USG	UNITED STATES GOVERNMENT
W/	WITH

DRAWFORM REVISION: 24 JULY 2019



1. WORK INDICATED ON THIS DRAWING(S) MUST COMPLY WITH FEDERAL BUILDING CODES.
2. CONTRACTOR SHALL BE RESPONSIBLE FOR THE APPROPRIATE STANDARD OF CARE REGARDING THE HANDLING OF ANY MATERIALS FOUND TO BE HAZARDOUS AND THE PROTECTION OF PERSONS FROM EXPOSURE.
3. MOVE AND/OR PROTECT MATERIAL THAT IS TO REMAIN IN PLACE OR IS TO BE RELOCATED UNLESS OTHERWISE NOTED.
4. WHERE PIPES, CONDUITS, DUCTS, OR OTHER UTILITIES ARE REMOVED AND NOT REPLACED, PROVIDE MATERIALS TO FILL THE OPENING(S) THAT ARE SIMILAR TO ADJACENT CONSTRUCTION.
5. FIELD VERIFY DIMENSIONS AND EXISTING CONDITIONS BEFORE CONSTRUCTION
6. WORK ASSOCIATED WITH THIS PACKAGE (I.E. - DELIVERY DATES, INSTALLATION, RELOCATIONS, ETC) MUST BE COORDINATED THROUGH THE PROJECT MANAGER, ALIKA PAU, 252.464.8828.

EXISTING TO REMAIN

DEMOLITION

APPLIES TO THIS SHEET ONLY

- 1 DEMOLISH 6 INCH CONCRETE SLAB AND SUBGRADE TO PROVIDE VAULT INSTALLATION. SEE STRUCTURAL DRAWING SD101



1/4" = 1'-0"

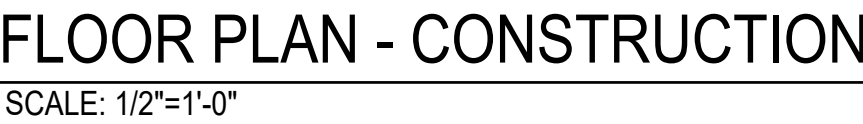
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7/18/2024
FINAL
DIVISION APPROVED
WHEN SIGNED

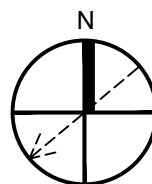
APPROVED	
FOR COMMANDER NAVFAC	
DES	S. FLICK
DRW	S. FLICK
CHK	C. HENRY
REVIEWED	A. PAU
APPROVED	E. LEWIS
NAVFAC PLAN PROJ NO: PE-22034	
SHOP: 93112	CAGE CODE 96916

DEPARTMENT OF THE NAVY
 U.S. INFRASTRUCTURE PLANNING & ENGINEERING BRANCH
 FLEET READINESS CENTER EAST
 NAVAL FACILITIES ENGINEERING COMMAND
 MOCS CHERRY POINT, NC
 U.S. MARINE CORPS AIR STATION, CHERRY POINT, NC
 DIRECT STATION
 FACILITY PREPARATION AND
 PROVIDE ULTRASONIC CLEANING SYSTEM
 FLOOR PLAN - DEMOLITION

DEPART	FACILITY	PROJECT
SCALE:		AS NOTED
CONSTR. CONTR. NO.		
#####		
NAVFAC DRAWING NO.		
12892411		
CP FEAD WO:		7357358
FRCE WO:		XZ62190
SHEET	9	OF 19
AD101		



SCALE: 1/2"=1'-0"



SHELF
168 X 36

1. USE CHEMICAL RESISTANCE RESINOUS COATING ON NEW SLAB. SEE SPECIFICATIONS

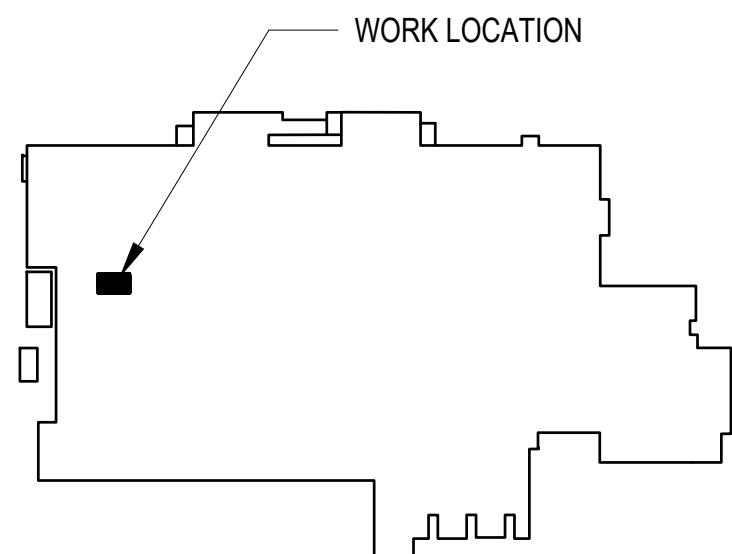
EXISTING TO REMAIN

CONSTRUCTION

FLOOR PAINT

APPLIES TO THIS SHEET ONLY

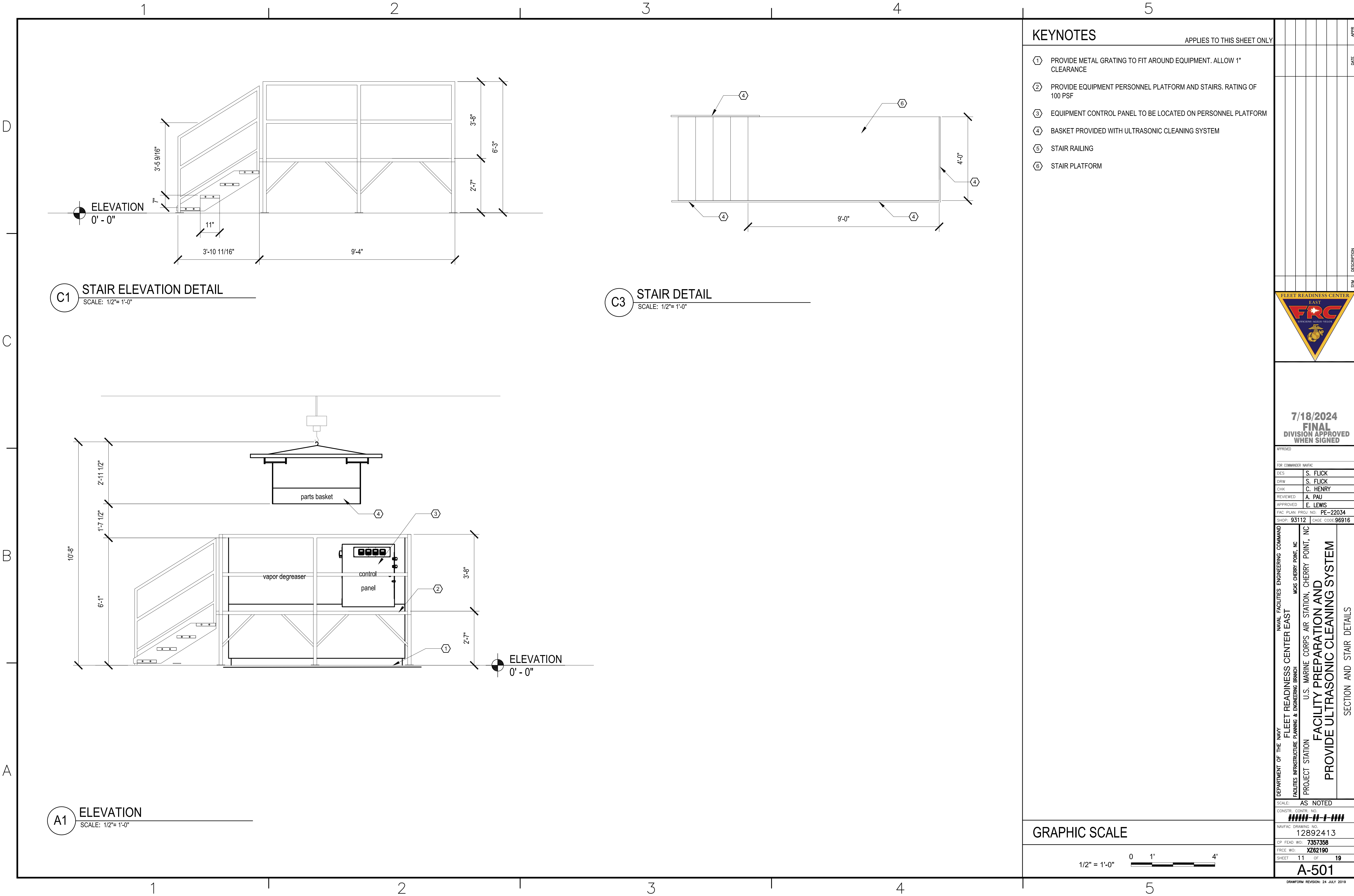
- ① PROVIDE ULTRASONIC CLEANING SYSTEM AND AUXILIARY EQUIPMENT PER SPECIFICATION SECTION 11 71 00. BASIS OF DESIGN (BOD): RELIANCE SPECIALTY PRODUCTS INC.; ULTRASONIC IMMERSION VAPOR DEGREASING SYSTEM; MODEL 6048-16LE-SPC.
- ② EQUIPMENT CONTROL PANEL. PROVIDED WITH ULTRASONIC CLEANER. LOCATE AS SHOWN
- ③ EQUIPMENT DRIP TRAY PROVIDED WITH ULTRASONIC CLEANER
- ④ PROVIDE FLOOR GRATING AROUND EQUIPMENT, FLUSH WITH FINISHED FLOOR.
- ⑤ PROVIDE EQUIPMENT PLATFORM AND STAIRS WITH RAILING. SEE DETAIL C1-501 AND C3-501
- ⑥ PROVIDE PERMANENT MOUNT POLYETHYLENE DELINEATOR POST WITH GRABBER/LOOPER TOP. SEE S-101.
- ⑦ PAINT RED TO MATCH EXISTING ISLE.
- ⑧ PAINT ISLE EDGE LINE YELLOW TO MATCH EXISTING.
- ⑨ PAINT NEW CONCRETE SLAB LIGHT GRAY TO MATCH EXISTING.



SCALE: NOT TO SCALE

1/2" = 1'-0"

A horizontal graphic scale bar with alternating black and white segments. It is marked with '0' at the left end, '1'' at the first segment boundary, and '4'' at the end of the fourth segment.[illegible]



GENERAL NOTES

UTILITIES AND EQUIPMENT SHOWN ARE BASED ON EXISTING PLANS AND FIELD OBSERVATION WITHOUT DEMOLITION. AFTER DEMOLITION, ANY CLARIFICATION REQUIRED TO DETERMINE SCOPE OF WORK MUST BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER.

DIMENSIONS SHOWN ARE AN APPROXIMATE. FIELD VERIFY ALL DIMENSIONS PRIOR TO ORDERING MATERIALS AND COMMENCING WORK .

REMOVAL OF ITEMS MUST INCLUDE ASSOCIATED SUPPORTS, ANCHOR BOLTS AND OTHER APPURTENANCES.

DEMOLISHED MATERIALS MUST BE REMOVED FROM SITE AND DISPOSED OFF BASE BY THE CONTRACTOR.

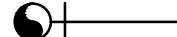
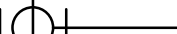
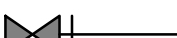
COORDINATE WORK SHOWN ON THE MECHANICAL DRAWINGS WITH ALL OTHER TRADES.

IT IS THE INTENTION OF THESE DRAWINGS TO COVER ALL WORK & MATERIAL FOR A COMPLETE FIRST CLASS INSTALLATION. ANY HARDWARE &/OR DEVICES NOT SPECIFICALLY MENTIONED OR SHOWN ON THE DRAWINGS, BUT NECESSARY FOR THE SATISFACTORY COMPLETION OF THE WORK (AS DETERMINED BY THE CONTRACTING OFFICER) MUST BE FURNISHED & INSTALLED BY THE CONTRACTOR AS PART OF HIS WORK.

IDENTIFY INSULATED AND UN-INSULATED PIPING IN ACCORDANCE WITH ANSI/ASME 13.1 TO INDICATE CONTENTS AND DIRECTION OF FLOW. INSTALL COMMERCIALY AVAILABLE SEMI-RIGID WRAP AROUND PIPE LABELS SUCH AS "DURALABEL" OR APPROVED EQUIVALENT. LOCATE LABELS AT 20 FEET INTERVALS, ON EACH SIDE OF WALL OR FLOOR PENETRATIONS, AT BRANCH TAKE-OFFS FROM MAINS AND AS CLOSE AS POSSIBLE TO EACH POINT OF SYSTEM USE. PAINT AND IDENTIFY PIPING AS FOLLOWS:

TEXT ON LABEL	BACKGROUND COLOR		LETTERS AND ARROW COLOR
SERVICE WATER	GREEN		WHITE
CHILLED WATER	GREEN		WHITE
OUTSIDE DIAMETER (INCLUDING INSULATION)	MINIMUM LENGTH OF LABEL	MINIMUM LETTER HEIGHT	FLOW ARROW WIDTH
1/2" TO 1 1/4"	8"	1/2"	1/2"
1 1/2" TO 2"	8"	1"	3/4"

LEGEND & ABBREVIATIONS

	EXISTING	AFF	ABOVE FINISHED FLOOR
	REMOVED OR PROVIDE	AFR	ABOVE FINISHED ROOF
	PIPE TURN UP	CHWR	CHILLED WATER RETURN
	PIPE TURN DOWN (DN)	CHWS	CHILLED WATER SUPPLY
	BALL VALVE	DW	DOMESTIC WATER
	GATE VALVE	ETR	EXISTING TO REMAIN
	PIPE INSULATION	MAU	MAKE UP AIR UNIT
	PIPE CONTINUES	TYP	TYPICAL
	TIE-IN POINT		
	PIPE TEE DOWN		
	REDUCER		
	BACKFLOW PREVENTER		

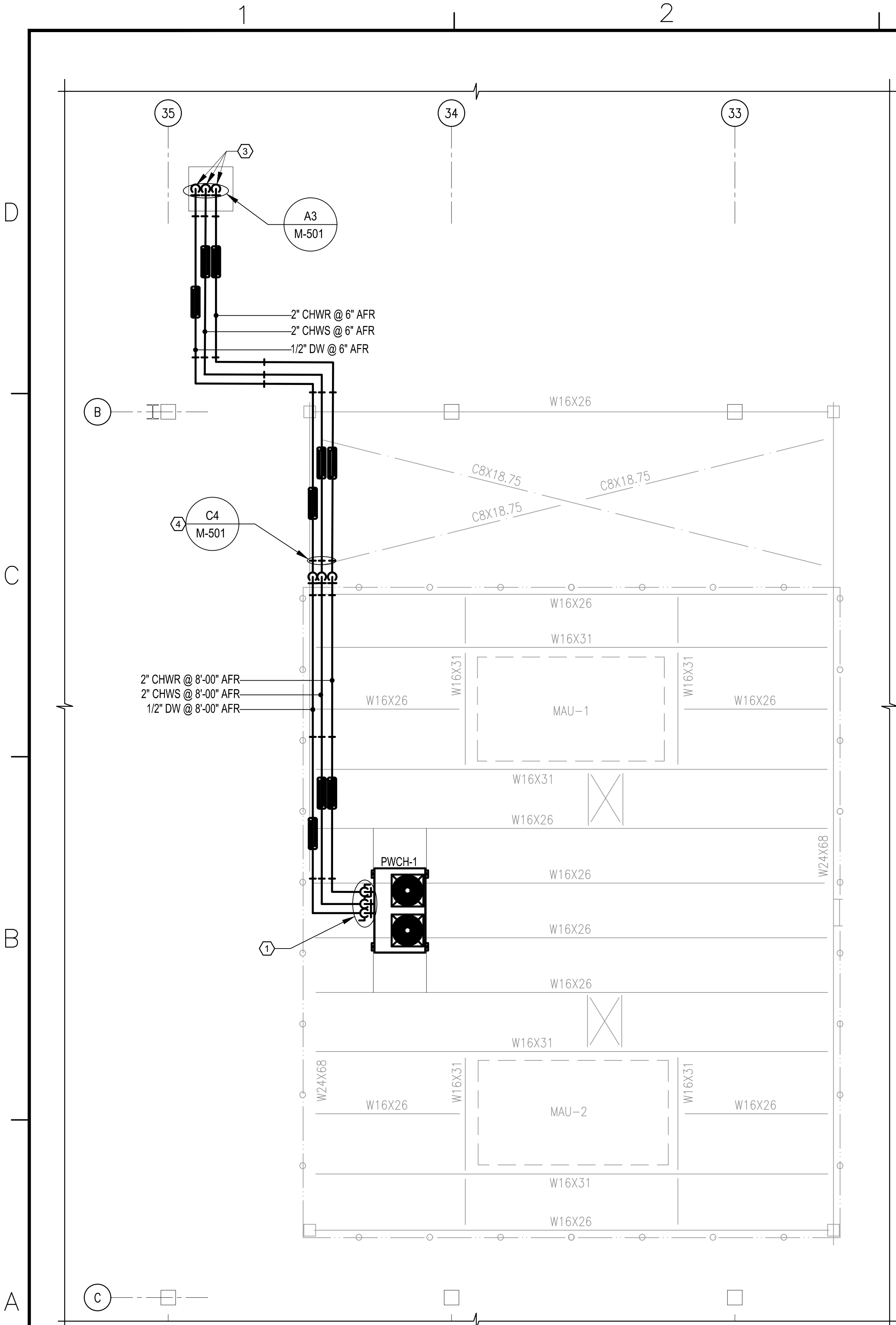
PROCESS WATER CHILLER PACKAGE SCHEDULE

TAG	GENERAL					COMPRESSOR				PUMP			RESERVOIR TANK		ELECTRICAL DATA			
	RATED CAPACITY (BTU/HR)	NOMINAL FLOW (GPM)	WATER LEAVING TEMPERATURE (MAX-°F)	DESIGN AMBIENT TEMPERATURE (°F)	OPERATING AMBIENT TEMPERATURES (°F)	QTY.	REFRIGERANT TYPE	REFRIGERANT (LBS)	PUMP (HP)	WATER PUMP QTY.	WATER PUMP HEAD (PSI)	PUMP (HP)	QTY.	VOLUME (GAL)	VOLTS	PHASE	FREQUENCY (Hz)	FULL LOAD CURRENT (AMP)
PWCH-1	90,000	25	65.0	95.0	20 - 95	1	R-134A	15	2	1	80	2	1	60	460	3	60	23

NOTES:

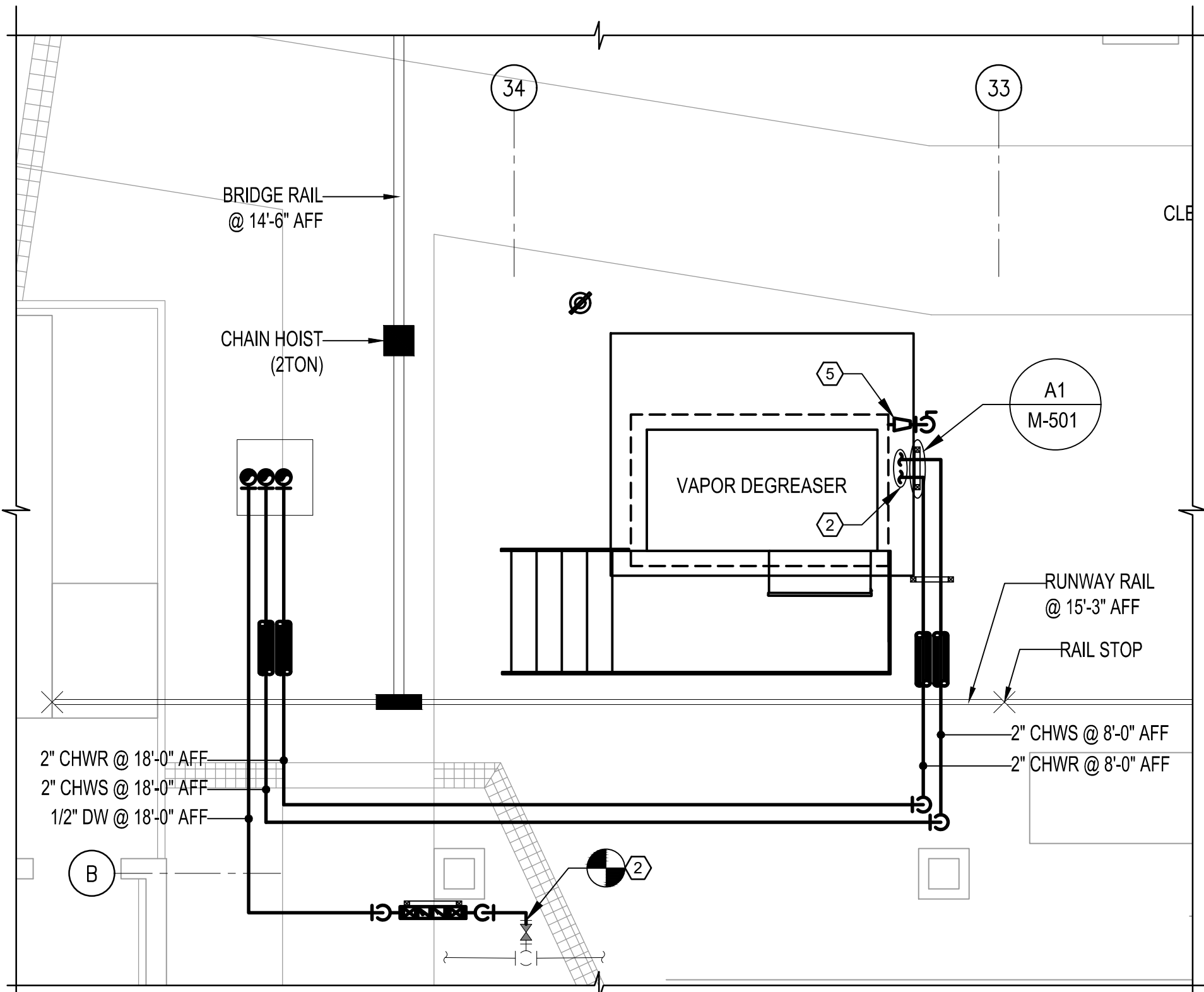
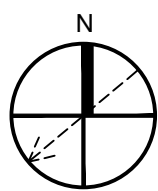
1. PACKAGED WATER CHILLER SYSTEM COMPLETE WITH:
 - PRESSURE PUMP UPGRADE
 - PRESSURE RELIEF BYPASS
 - AUTO WATER FILL
 - OUTDOOR RATED W/ SEASIDE COIL PROTECTION
 - INTEGRATED MICROPROCESSOR CONTROLS WITH REMOTE OPERATOR CONSOLE
2. SIMILAR OR EQUIVALENT TO KOOLANT KOOLERS S SERIES MODEL SVO-7500-M.

[illegible]



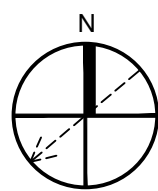
MECHANICAL ROOF PLAN - CONSTRUCTION

SCALE: 1/4"=1'-0"



MECHANICAL FLOOR PLAN - CONSTRUCTION

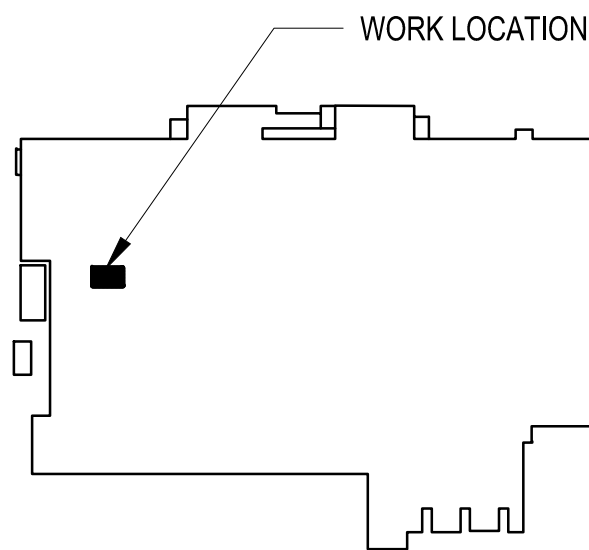
SCALE: 1/4"=1'-0"



KEYNOTES

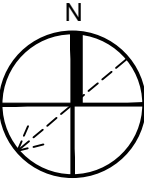
APPLIES TO THIS SHEET ONLY

1. CONNECT NEW CHILLED WATER RETURN/SUPPLY TO CHILLER. SEE M-701 FOR DETAIL.
2. CONNECT NEW CHILLED WATER RETURN/SUPPLY AND DOMESTIC WATER TO EQUIPMENT.
3. CHILLED WATER RETURN/SUPPLY AND DOMESTIC WATER THROUGH EXISTING ROOF HATCH. SEE A3, M-501 FOR DETAIL
4. TYPICAL FOR ROOF PIPE SUPPORTS.
5. PROVIDE 1-1/2" TO 3/4" REDUCER AND DRAIN VALVE WITH CAP TO EQUIPMENT DRAIN.
6. CONNECT TO EXISTING WATER SUPPLY.



B133 WORK LOCATION

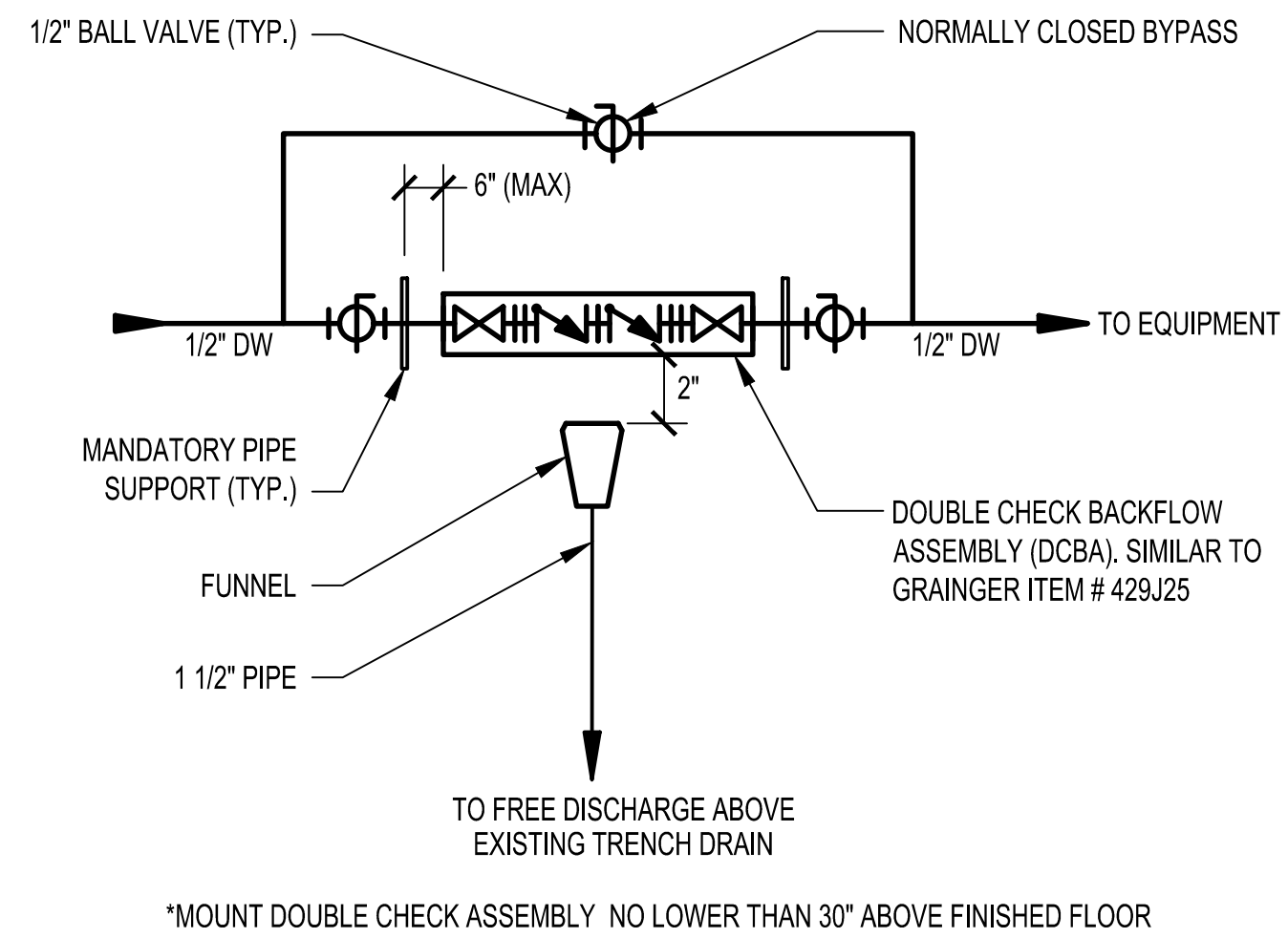
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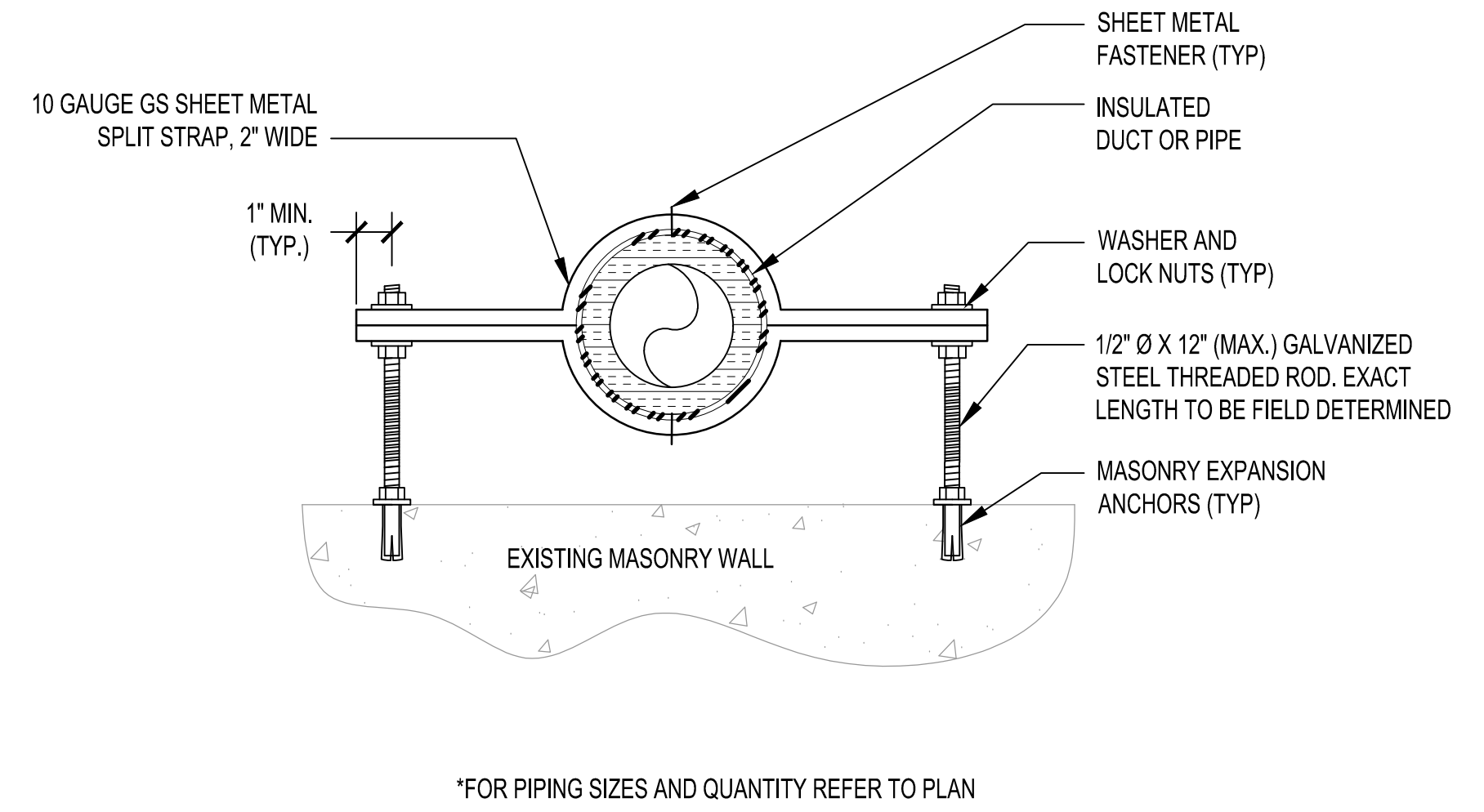
GRAPHIC SCALE



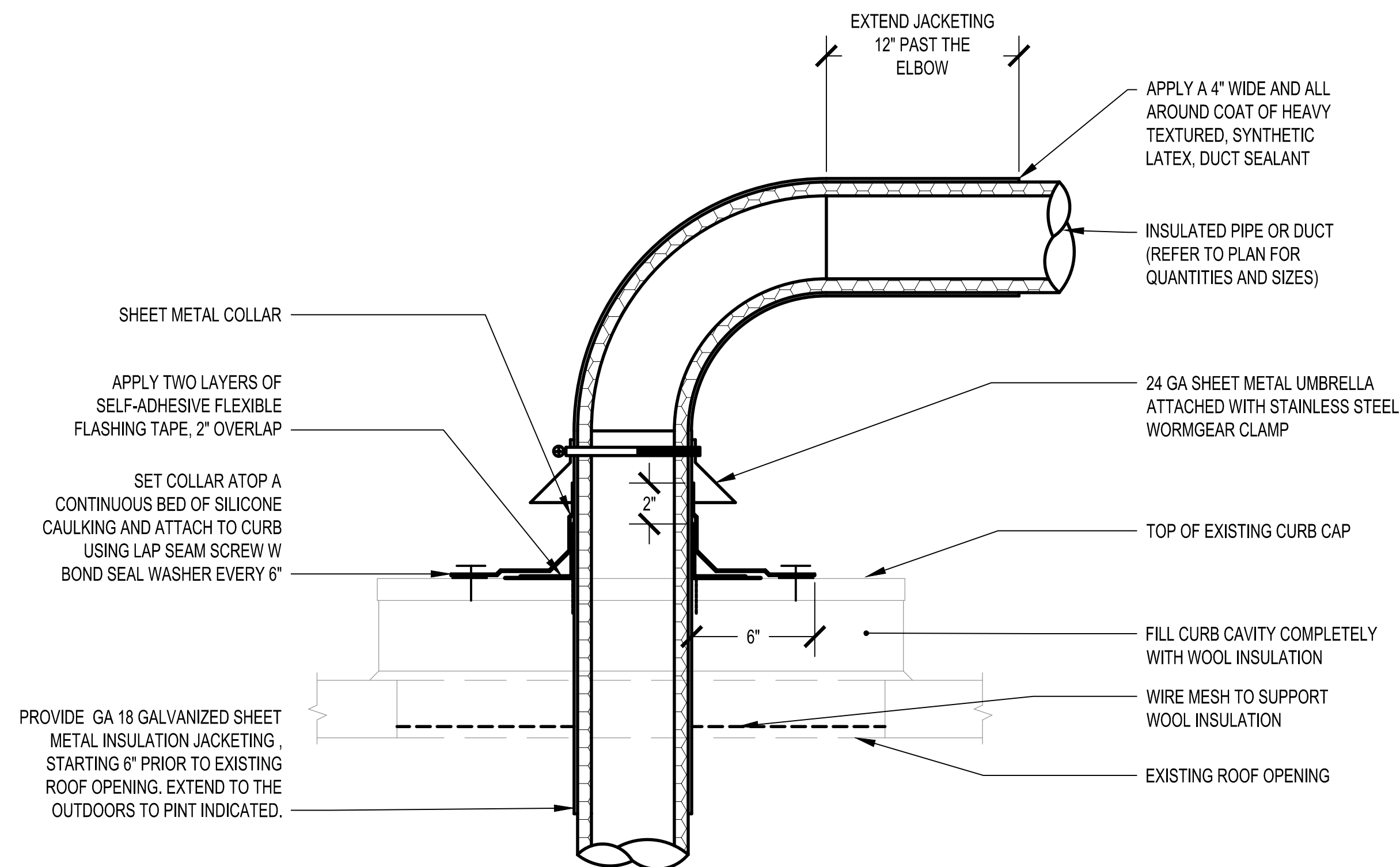
FLEET READINESS CENTER		DATE	APPR
U.S. MARINE CORPS AIR STATION, CHERRY POINT, NC		DESCRIPTION	SYN
FACILITY PREPARATION AND PROVIDE ULTRASONIC CLEANING SYSTEM		DESCRIPTION	SYN
MECHANICAL FLOOR & ROOF PLAN - CONSTRUCTION		DESCRIPTION	SYN
7/18/2024		DATE	APPR
FINAL		DATE	APPR
DIVISION APPROVED		DATE	APPR
WHEN SIGNED		DATE	APPR
FOR COMMANDER NAVFAC		DATE	APPR
DES		DATE	APPR
DRW		DATE	APPR
CHK		DATE	APPR
REVIEWED		DATE	APPR
APPROVED		DATE	APPR
FAC PLAN PROJ NO: PE-22034		DATE	APPR
SHOP: 93112		DATE	APPR
CAGE CODE: 96916		DATE	APPR
NAVAL FACILITIES ENGINEERING COMMAND		DATE	APPR
FLEET READINESS CENTER EAST		DATE	APPR
FACILITIES INFRASTRUCTURE PLANNING & ENGINEERING BRANCH		DATE	APPR
PROJECT STATION		DATE	APPR
SCALE: AS NOTED		DATE	APPR
CONSTR. CONTR. NO.		DATE	APPR
NAVFAC DRAWING NO.		DATE	APPR
12892415		DATE	APPR
CP FEAD WD: 7357358		DATE	APPR
TRCE WD: XZ62190		DATE	APPR
SHEET 13 OF 19		DATE	APPR
M-101		DATE	APPR
DRAWING REVISION: 24 JULY 2019		DATE	APPR



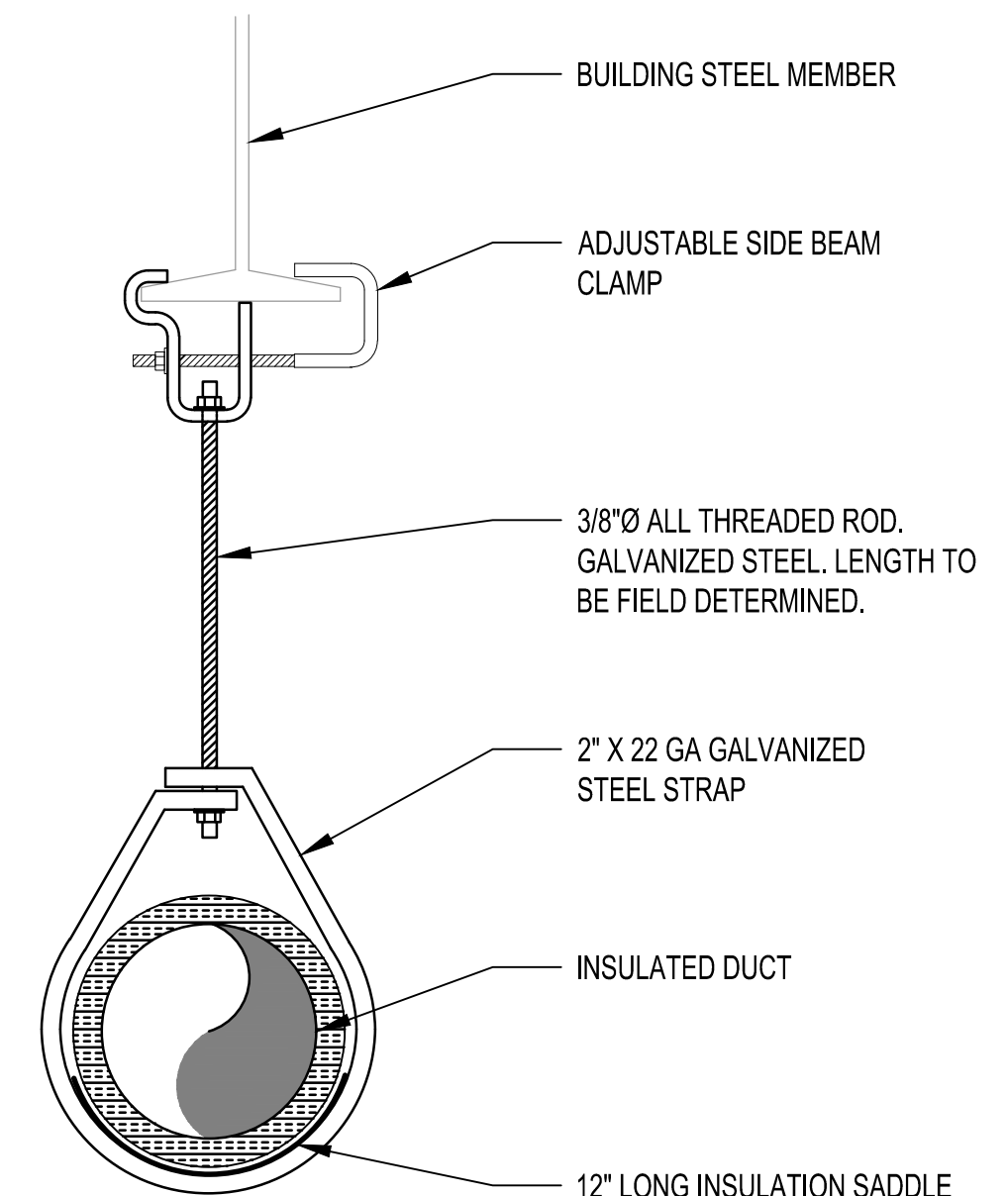
C2 DETAIL - DOUBLE BACK FLOW PREVENTER
NOT TO SCALE



C5 DETAIL -WALL SUPPORT
NOT TO SCALE



A3 **DETAIL - CURB PENETRATION**
NOT TO SCALE

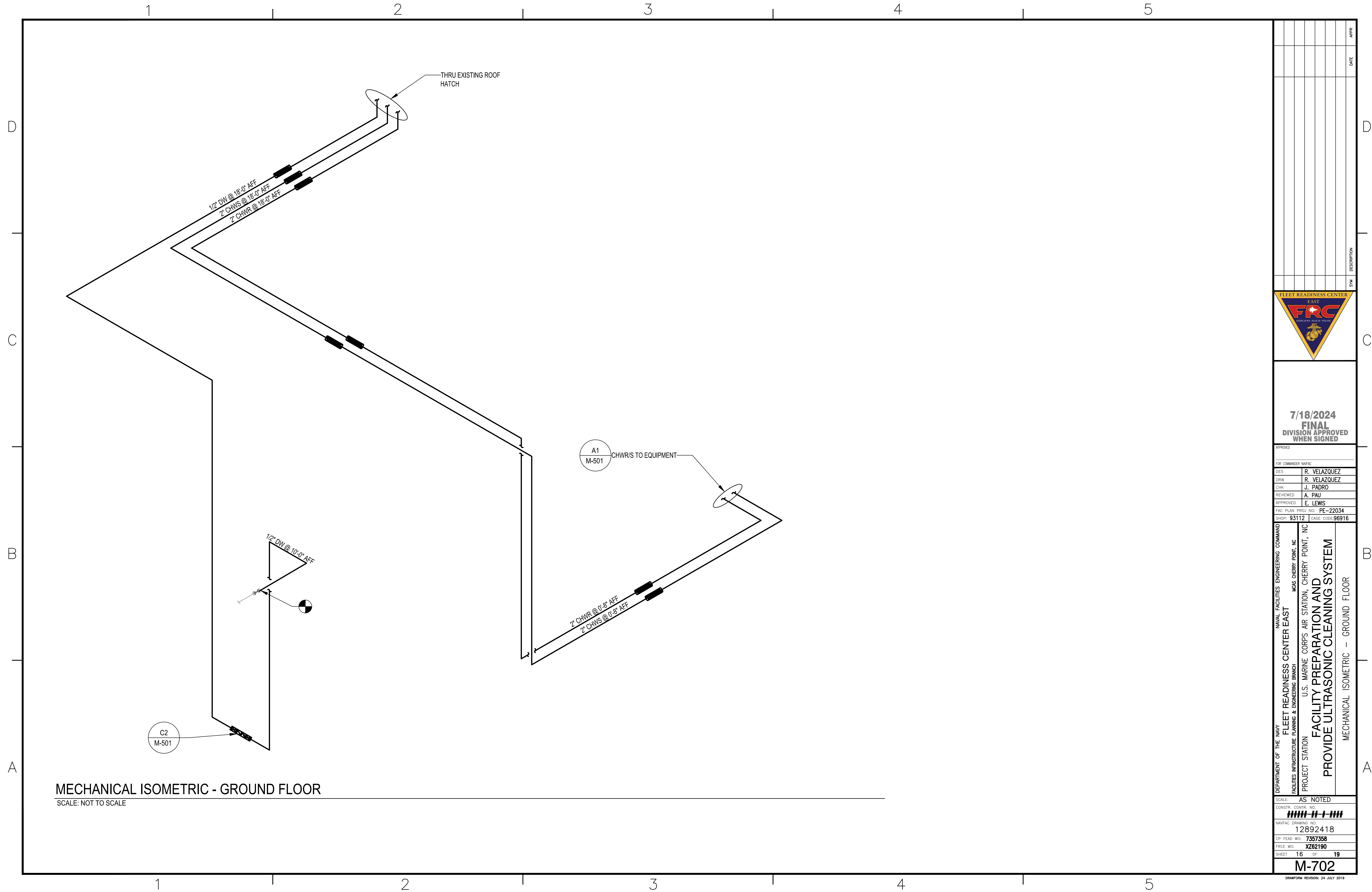


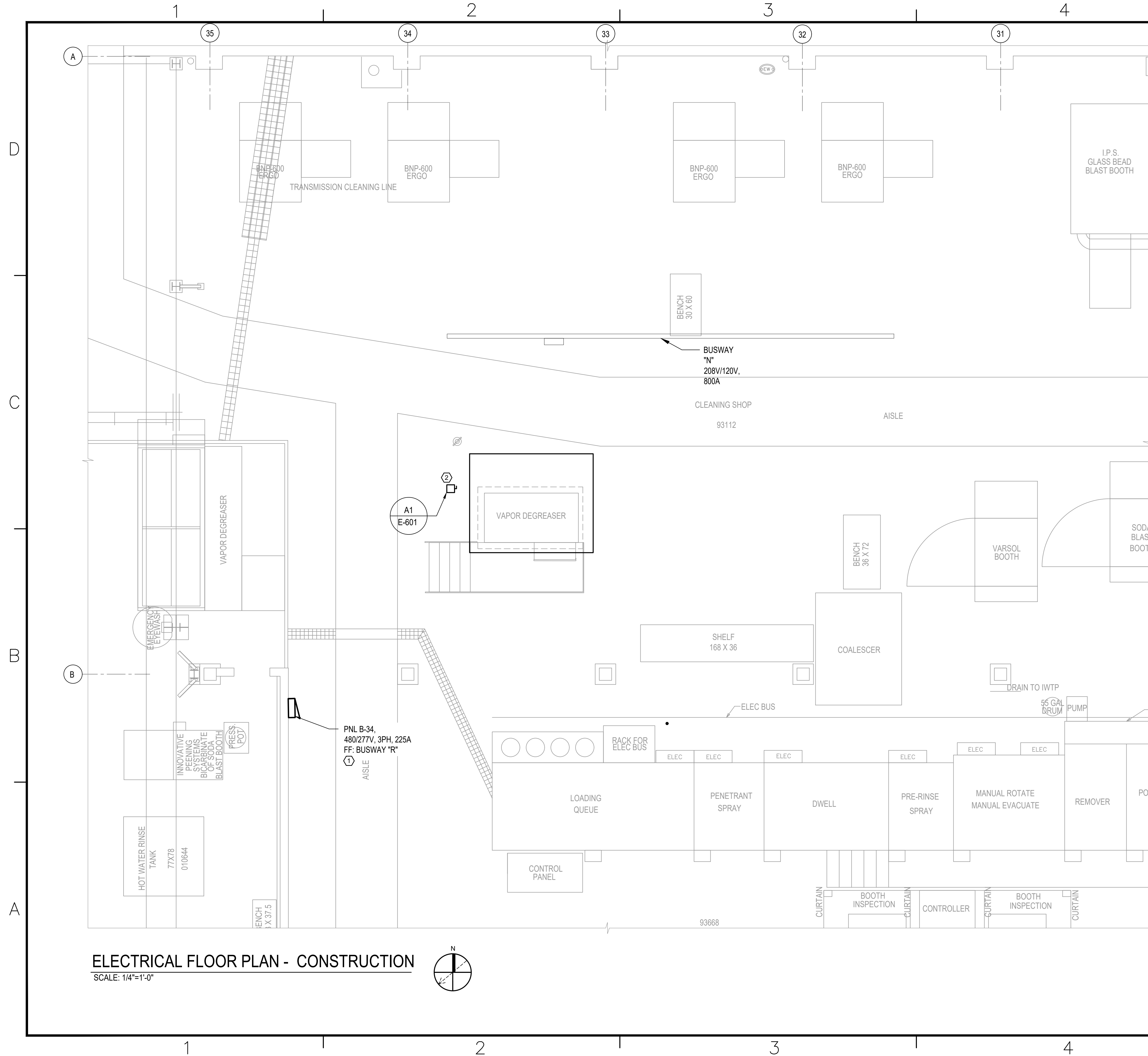
NOTES:

- PROVIDE SUPPORTS WITHIN 18" OF EVERY TURN
- 12'-0" MAXIMUM HANGERS SPACING

A5 **DETAIL - PIPE SUPPORT**
NOT TO SCALE

[illegible]





GENERAL SHEET NOTES

1. MOVE AND/OR PROTECT MATERIAL THAT IS TO REMAIN IN PLACE OR IS TO BE RELOCATED, UNLESS OTHERWISE NOTED.
2. FIELD VERIFY DIMENSIONS AND EXISTING CONDITIONS BEFORE CONSTRUCTION

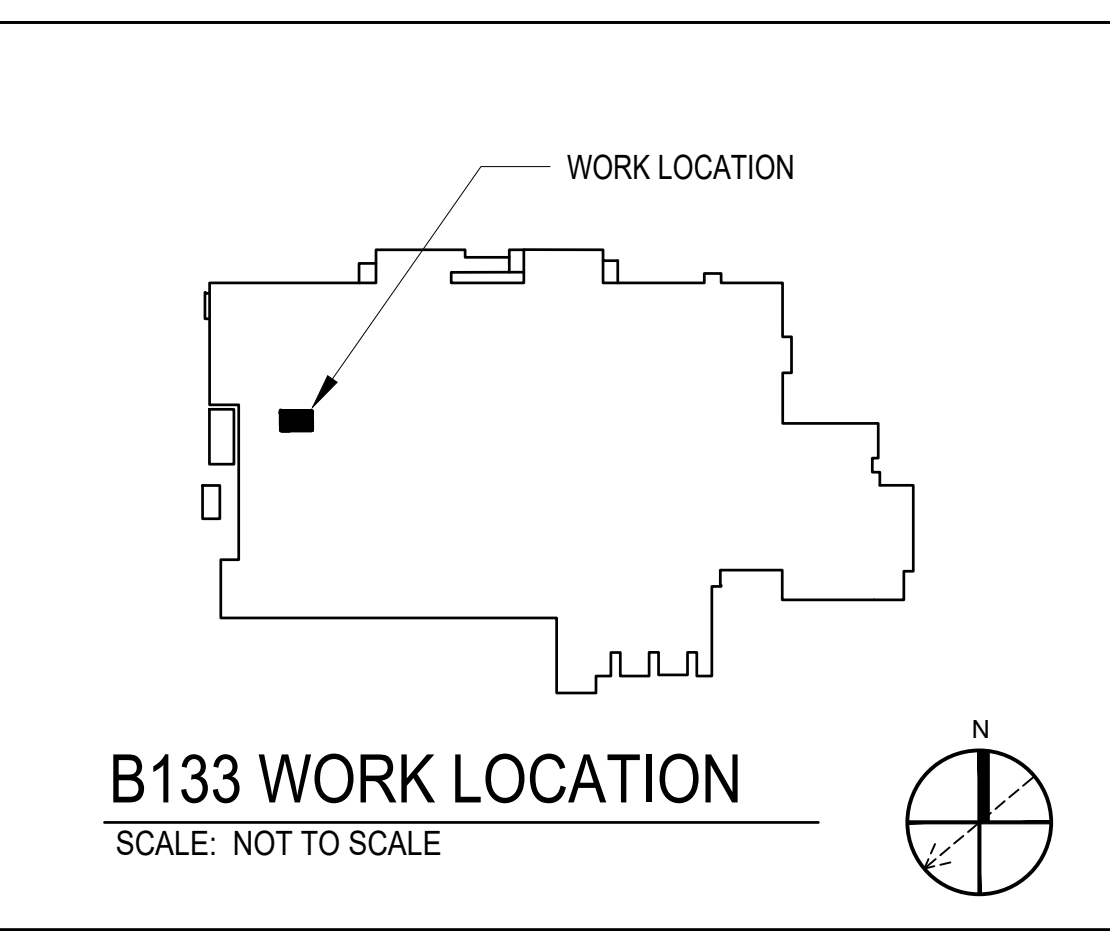
LEGEND AND ABBREVIATIONS

	EXISTING TO REMAIN	A AMPERE
	CONSTRUCTION	PNL PANELBOARD
	DISCONNECT SWITCH	PH PHASE
	PANELBOARD	V VOLTAGE
		AIC AMPERE INTERRUPTING CURRENT
		FF FED FROM

KEYNOTES

APPLIES TO THIS SHEET ONLY

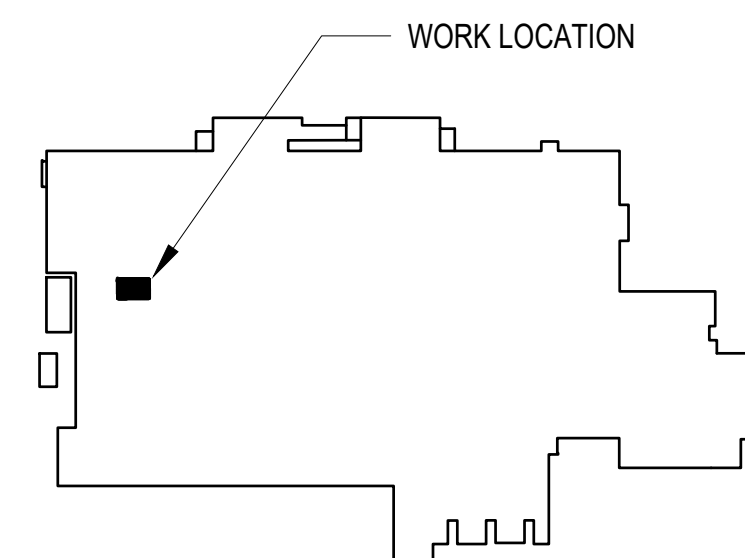
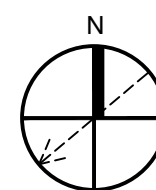
- ① PROVIDE ONE 480V, 3 PH, 30 A BREAKER TO FEED THE CHILLER. PROVIDE ONE 480V, 3 PH BREAKER SIZED, PER MANUFACTURERS REQUIREMENTS TO FEED THE VAPOR DEGREASER. BREAKERS MUST MATCH THE TYPE AND AIC RATING OF THE PANEL.
- ② PROVIDE A 600V, HEAVY DUTY, NEMA 1, FUSED DISCONNECT SWITCH. BRANCH CIRCUIT MUST BE FED FROM PANEL B-34 CIRCUIT 10/12/14. PROVIDE THHN/THWN WIRING IN ELECTRIC METALLIC TUBE FROM PANEL B-34 TO THE FUSED DISCONNECT SWITCH. MOUNT THE DISCONNECT SWITCH TO UNI-STRUT AS DETAILED ON A1/E-601. SIZE OF THE DISCONNECT, FUSING, WIRING AND CONDUIT WILL BE DETERMINED BASED ON ACTUAL EQUIPMENT REQUIREMENTS. THE CURRENT CANNOT EXCEED A MAXIMUM OF 80 A AT 480 V. MINIMUM SIZE CONDUIT 3/4". MINIMUM SIZE WIRE #12 FOR ALL NON CONTROL WIRING. PROVIDE AND INSTALL THE LABEL SHOWN IN DETAIL D4 ON SHEET E-601 AND THE ARC FLASH STICKER SHOWN ON E-601 ON THE DISCONNECT SWITCH.



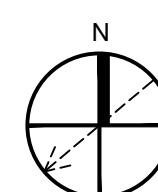
GRAPHIC SCALE

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ELECTRICAL ROOF PLAN - CONSTRUCTION



B133 WORK LOCATION
SCALE: NOT TO SCALE



GRAPHIC SCALE



GENERAL SHEET NOTES

1. WHERE PIPES, CONDUITS, DUCTS, OR OTHER UTILITIES ARE REMOVED AND NOT REPLACED, PROVIDE MATERIALS TO FILL THE OPENING(S) THAT ARE SIMILAR TO ADJACENT CONSTRUCTION.
2. FIELD VERIFY DIMENSIONS AND EXISTING CONDITIONS BEFORE CONSTRUCTION
3. WORK ASSOCIATED WITH THIS PACKAGE (I.E. - DELIVERY DATES, INSTALLATION, RELOCATIONS, ETC) MUST BE COORDINATED THROUGH THE PROJECT MANAGER, JOHN ALIKA PAU, 252.464.8828.

LEGEND AND ABBREVIATIONS

□ FUSED ELECTRIC DISCONNECT
A AMPERE
PNL PANELBOARD
PH PHASE
V VOLTAGE
AIC AMPERE INTERRUPTING CURRENT
FF FED FROM

KEYNOTES

APPLIES TO THIS SHEET ONLY

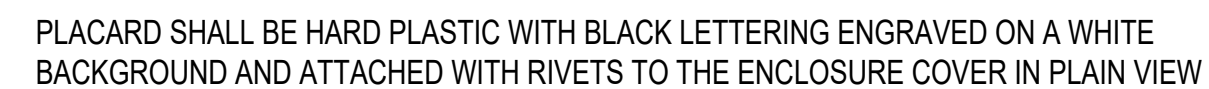
- ① PROVIDE A HEAVY DUTY, 600V, NEMA 3R, 30 A FUSED DISCONNECT SWITCH. BRANCH CIRCUIT MUST BE FED FROM PANEL B-34, CIRCUIT 13/15/17. PROVIDE 3 #10 THHN/THWN IN 3/4" RIGID CONDUIT FROM PANEL B-34 TO THE SWITCH. FOLLOW THE CHILLER PIPING TO THE DISCONNECT SWITCH. MOUNT DISCONNECT SWITCH TO UNI-STRUT AS DETAILED ON A1/E-601. PROVIDE AND INSTALL THE LABEL SHOWN IN DETAIL D4 ON SHEET E-601 AND THE ARC FLASH STICKER SHOWN ON E-601 ON THE DISCONNECT SWITCH.

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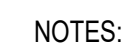
* - LOCK ON CIRCUIT BREAKER



SCALE: NOT TO SCALE



SCALE: NOT TO SCALE



1. PROVIDE SELF-ADHESIVE VINYL LABEL TO AFFIX TO ELECTRICAL EQUIPMENT TO WARN OF ARC FLASH HAZARDS.
2. THE LABEL FORMAT AND TEXT MUST BE IN ACCORDANCE WITH THE FIGURE.
3. THE LABEL MUST BE LOCATED ON THE EQUIPMENT TO BE CLEARLY VISIBLE TO QUALIFIED PERSONS BEFORE EXAMINATION, ADJUSTMENT, SERVICING, OR MAINTENANCE OF THE EQUIPMENT.
4. THE SIZE OF THE LABEL MUST BE MINIMUM:

EQUIPMENT TYPE	HEIGHT	WIDTH
INDOOR	2"	3"
OUTDOOR	3"	4.5"

5. A DOWNLOADABLE WINDOWS METAFILE IS AVAILABLE ON THE WHOLE BUILDING DESIGN GUIDE WEBSITE (WWW.WDBG.ORG) FOR USE IN A LABEL MAKING MACHINE.
- A. THE FILE IS LOCATED ON THE "NAVFAC CADD DETAILS" PAGE. TO NAVIGATE TO THIS LOCATION, FOLLOW: [HOME](#) > [DOCUMENTS & REFERENCES](#) > [CCB](#) > [CADD LIBRARY](#) > [NAVFAC CADD RESOURCES](#) > [NAVFAC CADD DETAILS](#).
 - B. ALTERNATIVELY, TYPE IN THE FOLLOWING ADDRESS IN INTERNET EXPLORER:
[HTTP://WWW.WDBG.ORG/CCB/BROWSE_CAT.PHP?C=232](http://WWW.WDBG.ORG/CCB/BROWSE_CAT.PHP?C=232)

SKETCH DATE

APRIL 2015

	STYLE
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AF-1

DRAWFORM REVISION: 24 JULY 2019